Asymptotic in Statistics

Lecture Notes for Stat522B

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Course Outline

A number of asymptotic results in statistics will be presented: concepts of statistic order, the classical law of large numbers and central limit theorem; the large sample behaviour of the empirical distribution and sample quantiles.

Prerequisite: Stat 460/560 or permission of the instructor.

Topics:

- Review of probability theory, probability inequalities.
- Modes of convergence, stochastic order, laws of large numbers.
- Results on asymptotic normality.
- Empirical distribution, moments and quartiles
- Smoothing method
- Asymptotic Results in Finite Mixture Models

Assessment: Students will be expected to work on 20 assignment problems plus a research report on a topic of their own choice.

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Chapter 1

Brief preparation in probability theory

1.1 Measure and measurable space

Measure theory is motivated by the desire of measuring the length, area or volumn of subsets in a space Ω under consideration. However, unless Ω is finite, the number of possible subsets of Ω is very large. In most cases, it is not possible to define a measure so that it has some desirable properties and it is consistent with common notions of area and volume.

Consider the one-dimensional Euclid space $\mathscr{R}$ consists of all real numbers and suppose that we want to give a length measurement to each subset of $\mathscr{R}$. For an ordinary interval $(a,b]$ with $b > a$, it is natural to define its length as

$$\mu((a,b]) = b - a,$$

where μ is the notation for measuring the length of a set. Let $I_i = (a_i, b_i]$ and $A = \cup I_i$ and suppose $a_i \leq b_i < a_{i+1}$ for all $i = 1, 2, \ldots$. It is natural to require μ to have the property such that

$$\mu(A) = \sum_{i=1}^{\infty} (b_i - a_i).$$

That is, we are imposing a rule on measuring the length of the subsets of $\mathscr{R}$.

Naturally, if the lengths of A_i, $i = 1, 2, \ldots$ have been defined, we want

$$\mu(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mu(A_i), \tag{1.1}$$

when A_i are mutually exclusive.

The above discussion shows that a measure might be introduced by first assigning measurements to simple subsets, and then be extended by applying the additive rule (1.1) to assign measurements to more complex subsets. Unfortunately, this procedure often does not extend the domain of the measure to all possible subsets of Ω. Instead, we can identify the maximum collection of subsets that a measure can be extended to. This collection of sets is closed under countable union. The notion of σ-algebra seems to be the result of such a consideration.

Definition 1.1 *Let Ω be a space under consideration. A class of subsets $\mathscr{F}$ is called a σ-algebra if it satisfies the following three conditions:*

(1) The empty set $\emptyset \in \mathscr{F}$;

(2) If $A \in \mathscr{F}$, then $A^c \in \mathscr{F}$;

(3) If $A_i \in \mathscr{F}$, $i = 1, 2, \ldots$, then their union $\cup_{i=1}^{\infty} A_i \in \mathscr{F}$.

Note that the property (3) is only applicable to countable number of sets. When $\Omega = \mathscr{R}$ and $\mathscr{F}$ contains all intervals, then the smallest possible σ-algebra for $\mathscr{F}$ is called Borel σ-algebra and all the sets in $\mathscr{F}$ are called Borel sets. We denote the Borel σ-algebra as $\mathscr{B}$. Even though not every subset of real numbers is a Borel set, statisticians rarely have to consider non-Borel sets in their research. As a side remark, the domain of a measure on R such that $\mu((a,b]) = b - a$, can be extended beyond Borel σ-algebra, for instance, Lesbegues algebra.

When a space Ω is equipped with a σ-algebra $\mathscr{F}$, we call $(\Omega, \mathscr{F})$ a measurable space: it has the potential to be equipped with a measure. A measure is formally defined as a set function on $\mathscr{F}$ with some properties.

Definition 1.2 *Let $(\Omega, \mathscr{F})$ be a measureable space. A set function μ defined on $\mathscr{F}$ is a measure if it satisfies the following three properties.*

(1) For any $A \in \mathscr{F}$, $\mu(A) \geq 0$;

(2) The empty set $\emptyset$ has 0 measure;

(3) It is countably additive:

$$\mu(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mu(A_i)$$

when A_i are mutually exclusive.

We have to restrict the additivity to countable number of sets. This restriction results in a strange fact in probability theory. If a random variable is continuous, then the probability that this random variable takes any specific real value is zero. At the same time, that chance for it to fall into some interval (which is made of individual values) can be larger than 0. The definition of a measure disallows adding up probabilities over all the real values in the interval to form the probability of the interval.

In measure theory, the measure of a subset is allowed to be infinity. We assume that $\infty + \infty = \infty$ and so on. If we let $\mu(A) = \infty$ for all non-empty set A, this set function satisfies the conditions for a measure. Such measures is probably not useful. Even if some sets possessing infinite measure, we would like to have a sequence of mutually exclusive sets such that every one of them have finite measure, and their union covers the whole space. We call this kind of measure σ-finite. Naturally, σ-finite measures have many other mathematical properties that are convenient in applications.

When a space is equipped with a σ-algebra $\mathscr{F}$, the sets in $\mathscr{F}$ have the potential to be measured. Hence, we have a measurable space $(\Omega, \mathscr{F})$. After a measure ν is actually assigned, we obtain a measure space $(\Omega, \mathscr{F}, \nu)$.

1.2 Probability measure and random variables

To a mathematician, a probability measure P is merely a specific measure: it assigns measure 1 to the whole space. The whole space is now called the sample space which denotes the set of all possible outcomes of an experiment. Individual possible outcomes are called sample points. For theoretical discussion, a specific experimental setup is redundant in the probability theory. In fact, we do not mention the sample space at all.

In statistics, the focus is on functions defined on the sample space Ω, and these functions are called random variables. Let X be a randon variable. The desire of

computing the probability of $\{\omega : X(\omega) \in B\}$ for a Borel set B makes it necessary for $\{\omega : X(\omega) \in B\} \in \mathscr{F}$. These considerations motive the definition of a random variable.

Definition 1.3 *A random variable is a real valued function on the probability* $(\Omega, \mathscr{F}, P)$ *such that* $\{\omega : X(\omega) \in B\} \in \mathscr{F}$ *for all Borel sets B.*

In plain words, random variables are $\mathscr{F}$-measurable functions.

Interestingly, this definition rules out the possibility for X to take infinity as its value and implies the cumulative distribution function defined as

$$F(x) = P(X \leq x)$$

has limit 1 when $x \to \infty$. For one-dimensional function $F(x)$, it is a cumulative distribution function of some random variable if and only if

1. $\lim_{x \to -\infty} F(x) = 0$; $\lim_{x \to \infty} F(x) = 1$.
2. $F(x)$ is a non-decreasing, right continuous function.

Note also that with each random variable defined, we could define a corresponding probability measure P_X on the real space such that

$$P_X(B) = P(X \in B).$$

We have hence obtained an induced measure on R. At the same time, the collection of sets $X \in B$ is also a σ-algebra. We call it $\sigma(X)$ which is a sub-σ-algebra of $\mathscr{F}$.

Definition 1.4 *Let X be a random variable on a probability space* $(\Omega, \mathscr{F}, P)$. *We define* $\sigma(X)$ *to be the smallest* σ*-algebra such that*

$$\{X \in B\} \in \sigma(X)$$

for all $B \in \mathscr{B}$.

It is seen that sum of two random variables is also a random variable. All commonly used functions of random variables are also random variables. That is, they remain $\mathscr{F}$-measurable.

The rigorous definitions of integration and expectation are involved. Let us assume that for a measurable function $f(\cdot) \geq 0$ on a measure space $(\Omega, \mathscr{F}, \nu)$, a simple definition of the integration

$$\int f(\cdot) d\nu$$

is available. A general function f can be written as $f^+ - f^-$, the difference between its positive and negative parts. The integration of this function is the difference between two separate integrations

$$\int f d\nu = \int f^+ d\nu - \int f^- d\nu$$

unless we are in the situation of $\infty - \infty$. In this case, the integration is said not exist. The expectation of a function of a random variable X is simply

$$\int f(X(\cdot)) dP = \int f(\cdot) dP_X$$

banning $\infty - \infty$. Note that the integrations on two sides are with respect to two different measures. The above equality was joked as un-conscience statistician's law. In mathematics, it is called the change-of-variable formula.

The integration taught in undergraduate calculus courses are called Riemann integration. Most properties of Riemann integration remain valid for this measure-theory-based integration. The new integration makes more functions integrable. Under the new definition (even though we did not really give one), it becomes unnecessary to separately define the expectation of continuous random variables and the expectation of discrete random variables. Without a unified definition, the commonly accepted formulas such as

$$\mathbb{E}(X+Y) = \mathbb{E}(X) + \mathbb{E}(Y)$$

are unprovable.

The concept of Radon-Nikodym derivative is hard to many, but it is handy for example when we have to work with discrete and continuous random variables on the same platform. Suppose ν and λ are two σ-finite measures on the measurable space $(\Omega, \mathscr{F})$. We say λ is dominated by ν if for any $\mathscr{F}$ measurable set A, $\nu(A) = 0$ implies $\lambda(A) = 0$. We use notation $\lambda << \nu$. Note that this definition is σ-algebra $\mathscr{F}$ dependent.

The famous Radon-Nikodym Theorem is as follows.

Theorem 1.1 *Let ν and λ be two σ-finite meausres on $(\Omega, \mathscr{F})$. If $\lambda << \nu$, then there exists a non-negative $\mathscr{F}$-measurable function f such that*

$$\lambda(A) = \int_A f d\nu$$

for all $A \in \mathscr{F}$.

We call f the Radon-Nikodym derivative of λ with respect to ν.

If λ is a probability measure, then f can be chosen to be non-negative, and it is called the density function of λ with respect to ν. The commonly referred probability density function is the derivative of the cumulative distribution function of absolutely continuous random variables with respect to Lesbesgues measure. The probability mass function is the derivative of the cumulative distribution function of integer valued random variables with respect to counting measure. One such example is the probability mass function of the Poisson distribution.

Essentially, a measure assigns a non-negative value to every member in the σ-fields it is designed on and possesses some properties. Lesbegues measure assigns a value of every interval equaling to its length. The value of every other set in the σ-field is derived from the rules (properties) for a measure. In comparison, under the counting measure we often do not explicitly define, each set with single integer has measure 1. Any set with finite number of integers has measure equaling the number of integers it contains.

1.3 Conditional expectation

The concept of expectation is developed as theoretical average size of a random variable. The word "conditional" has a meaning tied more tightly with probability. When we focus on a subset of the sample space and rescale the probability on this event to 1, we get a conditional probability measure. In elementary probability theory, the conditional expectation is also the average size of a random variable where we only examine its behaviour when its outcome is in a pre-specified subset of the sample space.

To understand the advanced notion of conditional expectation, we start with an indicator random variable. By taking values 1 and 0 only, an indicator random

variable I_A divides the sample space into two pieces: A and A^c. The conditional expectation of a random variable X given $I_A = 1$ is the average size of X when A occurs. Similarly, the conditional expectation of X given $I_A = 0$ is the average size of X over A^c. Thus, random variable I_A partitions Ω into two pieces, and we compute the conditional expectation of X over each piece. We may use a random variable Y to cut the sample space into more pieces and compute conditional expectations over each. Consequently, the conditional expectation of X given Y becomes a function: it takes different values on different pieces of the sample space and the partition is created by the random variable Y.

If the random variable Y is not discrete, it does not partition the sample space neatly. A general random variable Y does not neatly partition the sample space into countable many mutually exclusive pieces. Computing average size of X given the size of Y is hard to image. At this moment, we may realize that the concept of σ-algebra can be helpful. In fact, with the concept of σ-algebra, we define the conditional expectation of X without the help of Y.

The conditional expectation is not given by a constructive definition, but a conceptual requirement on what properties it should have.

Definition 1.5 *The conditional expectation of a random variable X given a σ-algebra $\mathscr{A}$, $\mathbb{E}(X|\mathscr{A})$, is a $\mathscr{A}$-measurable function such that*

$$\int_A \mathbb{E}(X|\mathscr{A})dP = \int_A XdP$$

for every $A \in \mathscr{A}$.

If Y is a random variable, then we define $\mathbb{E}(X|Y)$ as $\mathbb{E}(X|\sigma(Y))$. It turns out such a function exists and is practically unique whenever the expectation of X exists. The conditional expectation defined in elementary probability theory does not contradict this definition. In view of this new definition, we must have $\mathbb{E}\{\mathbb{E}(X|Y)\} = \mathbb{E}(X)$. This formula is true by definition! We regret that the above definition is not too useful for computing $\mathbb{E}(X|Y)$ when given two random variables X and Y.

When working with conditional expectation under measure theory, we should remember that the conditional expectation is a random variable. It is regarded as non-random with respect to the σ-algebra in its conditional argument. Most

formulas in elementary probability theory have their measure theory versions. For example, we have

$$\mathbb{E}[g(X)h(Y)|Y] = h(Y)\mathbb{E}[g(X)|Y]$$

whenever the relevant quantities exist.

The definition of conditional probability can be derived from the conditional expectation. For any event A, we note that $P(A) = \mathbb{E}\{I_A\}$. Hence, we regard the conditional probability $P(A|B)$ as the the value of $\mathbb{E}\{I_A|I_B\}$ when the sample point $\omega \in B$. To take it to extreme, many probabilists advocate to forego the probability operation all together.

1.4 Independence

The probability theory becomes a discipline rather than a special case of the measure theory largely due to some special notions so dear to probabilistic concepts.

Definition 1.6 *Let $(\Omega, \mathscr{F}, P)$ be a probability space. Two events $A, B \in \mathscr{F}$ are independent if any only if $P(AB) = P(A)P(B)$.*

Let $\mathscr{F}_1$ and $\mathscr{F}_2$ be two sub-σ-algebras of $\mathscr{F}$. They are independent if any only if A is independent of B for all $A \in \mathscr{F}_1$ and $B \in \mathscr{F}_2$.

Let X and Y be two random variables. We say that X and Y are independent if and only if $\sigma(X)$ and $\sigma(Y)$ are independent of each other.

Conceptually, when A and B are two independent events, then $P(A|B) = P(A)$ by the definition in elementary probability theory textbooks. Yet one cannot replace $P(AB) = P(A)P(B)$ in the above independence definition by $P(A|B) = P(A)$. It becomes problematic when, for example, $P(B) = 0$.

Theorem 1.2 *Two random variables X and Y are independent if and only if*

$$P(X \le x, Y \le y) = P(X \le x)P(Y \le y) \tag{1.2}$$

for any real numbers x and y.

The generalization to a countable number of random variables can be done easily. A key notion is, pairwise independence is not sufficient for full independence.

1.5 Assignment problems

1. Let X be a random variable having Poisson distribution with mean $\mu = 1$, Y be a random variable having standard normal distribution and W be a random variable such that $P(W = 0) = P(W = 1) = 0.5$.

   Assume X, Y and W are independent. Construct a measure $\nu(\cdot)$ such that it dominates the probability measure induced by $WX + (1-W)Y$.

2. Let the space Ω be the set of all real numbers. Suppose a σ-algebra $\mathscr{F}$ contains all half intervals in the form of $(-\infty, x]$ for all real number x. Show that $\mathscr{F}$ contains all singleton set $\{x\}$.

3. Let $\mathscr{B}$ be the Borel σ-algebra on R and that Y is a random variable. Verify that
   $$\sigma(Y) = \{Y^{-1}(B) : B \in \mathscr{B}\}$$
   is a σ-algebra, where $Y^{-1}(B) = \{\omega : Y(\omega) \in B\}$.

4. From measurability point of view, show that if X and Y are two random variables, then $X + Y$, XY are also random variables.

   Give an example where X/Y is not a random variable if the definition in section 1.2 is rigorously interpreted.

5. Prove that if $F(x)$ is a cumulative distribution function of some random variable, then
   $$\lim_{x \to -\infty} F(x) = 0, \quad \lim_{x \to \infty} F(x) = 1.$$

6. Assume that $g(\cdot)$ is a measurable function and Y is a random variable. Assume both $\mathbb{E}(Y)$ and $\mathbb{E}\{g(Y)\}$ exist. Prove or disprove that
   $$\mathbb{E}\{g(Y)|Y\} = g(Y); \quad \mathbb{E}\{Y|g(Y)\} = Y.$$

7. Assume all relevant expectations exist. Show that
   $$\mathbb{E}[g(X)h(Y)|Y] = h(Y)\mathbb{E}[g(X)|Y]$$
   provide that both g and h are measurable functions. The equality may be interpreted as valid except on a measurable zero-probability event.

8. Define $\text{VAR}(X|Y) = \mathbb{E}[\{X - \mathbb{E}(X|Y)\}^2|Y]$. Show that

$$\text{VAR}(X) = \mathbb{E}\{\text{VAR}(X|Y)\} + \text{VAR}\{\mathbb{E}(X|Y)\}.$$

9. Prove Theorem 1.2.

10. Prove that if X and Y are independent random variables, and h and g are two measurable functions,

$$\mathbb{E}[h(X)g(Y)] = \mathbb{E}[h(X)]\mathbb{E}[g(Y)]$$

under the assumption that all expectations exist and finite.

11. Suppose X and Y are jointly normally distributed with means 0, variances 1, and correlation coefficient ρ.

Verify that $\mathbb{E}(X|Y) = \rho Y$.

Remark: rigorous proofs of some assignment problems may need some knowledges beyond what have been presented in this chapter. It is hard to clearly state what results should be assumed. Hence, we have to leave a big dose of ambiguity here. Nevertheless, these problems show that some commonly accepted results are not self-righteous. They are in fact rigorously established somewhere.

Chapter 2

Fundamentals in Asymptotic Theory

Other than a few classical results in mathematical statistics, the exact distributional property of a statistics or other random objects is often hard determine to the last details. A good approximation to the exact distribution is very useful in investigating the properties of various statistical procedures. In statistical applications, many observations, say n of them, from the same probability model/population are often *assumed* available. Good approximations are possible when the number of repeated observations is large. A theory developed for the situation where the number of observations is large forms the *Asymptotic Theory*.

In asymptotic theory, we work hard to find the limiting distribution of a random quantity sequence T_n as $n \to \infty$. Such results are sometime interesting for their own rights. In statistical applications, we do not really have the sample size n increases as time goes, much less where n increases unboundedly. If so, why should we care about the limit which is usually attained only when $n = \infty$? My answer is similar to the answer to the use of tangent line to replace a segment of smooth curve in mathematical analysis. If $f(x)$ is a smooth function at a neighborhood of $x = 0$, we have approximately

$$f(x) \approx f(0) + f'(0)x.$$

While the approximation may never be exact unless $x = 0$, we are comfortable to claim that if the approximation is precise enough at $x = 0.1$, it will be precise enough for $|x| \leq 0.1$. In asymptotic theory, if the limiting distribution approximates the finite sample distribution when $n = 100$ well enough, we are confident

that when $n > 100$, the approximation will likely be more accurate. In this situation, we are comfortable to use the limiting distribution in the place of the exact distribution for statistical inference.

In this chapter, we introduce some classical notion and results in limiting process.

2.1 Mode of convergence

Let $X_1, X_2, \ldots, X_n, \ldots$ be a sequence of random variables defined on a probability space with sample space Ω, σ-algebra $\mathscr{F}$, and probability measure P.

Recall that every random variable is a real valued function. Thus, a sequence of random variables is also a sequence of functions. At each sample point $\omega \in \Omega$, we have a sequence of real numbers:

$$X_1(\omega), X_2(\omega), \ldots.$$

For some ω, the limit of the above sequence may exist. For some other ω, the limit may not exist. Let $A \subset \Omega$ be the set of ω at which the above sequence converges. It can be shown that A is then measurable. Let X be a random variable such that for each $\omega \in A$, $X(\omega) = \lim_{n\to\infty} X_n(\omega)$.

Definition 2.1 Convergence almost surely: *If $P(A) = 1$, we say that $\{X_n\}_{n=1}^{\infty}$ converges almost surely to X. In notation, $X_n \xrightarrow{a.s.} X$.*

A minor point is that the limit X is unique up to a zero probability event under the conditions in the above definition. If another random variable Y differs from X by a zero probability event, then we also have $X_n \to Y$ almost surely. Proving the almost sure convergence of a random variable sequence is often hard. A weaker version of the convergence is much easier to establish.

Let X, $\{X_n\}_{n=1}^{\infty}$ be one and a sequence of random variables defined on a probability space. In weak version of convergence, we examine the probability of the difference $X - X_n$ being large.

Definition 2.2 Convergence in probability. *Suppose that for any $\delta > 0$,*

$$\lim_{n\to\infty} P\{|X_n - X| \geq \delta\} = 0.$$

Then we say that X_n converges to X in probability. In notation, $X_n \xrightarrow{p} X$.

Conceptually, the mode of almost sure convergence keeps track of the values of random variables at the same sample point on and on. It requires the convergence of $X(\omega)$ at almost all sample points. If you find "almost all sample points" is too tricky, simply interpret it as "all sample points" and you are not too far from the truth. The mode of convergence in probability requires that the event on which X_n and X differ more than a fixed amount shrinks in probability. This event is n dependent. It is one event when $n = 10$ and and it is another when $n = 11$ and so on. In other words, we have a moving target as n evolves when defining convergence in probability. Because of this, the convergence in probability does not imply convergence of $X_n(\omega)$ for any $\omega \in \Omega$. The following classical example is a vivid illustration of this point.

Example 2.1 *Let $\Omega = [0,1]$, the unit interval of real numbers. Let $\mathscr{F}$ be the classical Borel σ-algebra on $[0,1]$ and P be the uniform probability measure.*

For $m = 0,1,2,\ldots,$ and $k = 0,1,\ldots,2^m - 1$, let

$$X_{2^m+k}(\omega) = \begin{cases} 1 & \text{when } k < 2^m\omega \le (k+1); \\ 0 & \text{otherwise}. \end{cases}$$

In plain words, we have defined a sequence of random variables made of indicator functions on intervals of shrinking length 2^{-m}. Yet the union of every 2^m intervals completely cover the sample space $[0,1]$ as k goes from 1 to 2^m.

It is seen that

$$P(|X_n| > 0) \le 2^{-m}$$

where $m = \log n/\log 2 - 1$. Hence as $n \to \infty$, $P(|X_n| > 0) \to 0$. This implies $X_n \to 0$ in probability.

At the same time, the sequence

$$X_1(\omega), X_2(\omega), X_3(\omega), \ldots$$

contains infinity numbers of both 0 and 1 for any ω. Thus none of such sequence converge. In other words,

$$P(\{\omega : X_n(\omega) \text{ converges}\}) = 0.$$

Hence, X_n does not converge to 0 in the mode of "almost surely".

Due to the finiteness of the probability measure, if a sequence of random variables X_n converges to X almost surely, then X_n also converges to X in probability. If a sequence of random variables X_n converges to X in probability, X_n does not necessarily converge to X as show by the above example. However, X_n always has a subsequence X_{n_k} such that $X_{n_k} \to X$ almost surely.

The convergence in moment is another commonly employed concept. It is often not directly applied in statistical theory, but it is sometimes convenient to verify the convergence in moment. The convergence in moment implies the convergence in probability.

Definition 2.3 Convergence in moment. *Let $r > 0$ be a real number. If the rth absolute moment exists for all $\{X_n\}_{n=1}^{\infty}$ and X, and*

$$\lim_{n\to\infty} E\{|X_n - X|^r\} = 0,$$

then X_n converges to X in the rth moment.

By a well known inequality in probability theory, we can show the rth moment convergence implies the sth moment convergence when $0 < s < r$. In addition, it also implies the convergence in probability. Such a result can be established easily by using the following inequality.

Markov Inequality: Let X be a random variable and for some $r > 0$, $E|X|^r < \infty$. Then for any $\varepsilon > 0$, we have

$$P(|X| \geq \varepsilon) \leq \frac{E|X|^r}{\varepsilon^r}.$$

PROOF: It is easy to verify that

$$I(|X| \geq \varepsilon) \leq \frac{|X|^r}{\varepsilon^r}.$$

Taking expectation results in the inequality to be shown. ♢

When $r = 2$, the Markov inequality becomes Chebyshev's inequality:

$$P(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}$$

where $\mu = E(X)$ and $\sigma^2 = Var(X)$.

Example 2.2 *Suppose $X_n \to X$ in the rth moment for some $r > 0$. For any $\delta > 0$, we have*

$$P(|X_n - X| \geq \delta) \leq \frac{E|X_n - X|^r}{\delta^r}.$$

The right hand side converges to zero as $n \to \infty$ because of the moment convergence. Thus, we have shown that $X_n \to X$ in probability.

The reverse of this result is not true in general. For example, let X be a random variable with uniform distribution on [0, 1]. Define $X_n = X + nI(X < n^{-1})$ for $n = 1, 2, \ldots$. It is easy to show that $X_n \to X$ almost surely. However, $E|X_n - X| = 1$ which does not converge to zero. Hence, X_n does not converge to X in the first moment. If, however, there exists a nonrandom constant M such that $P(|X_n| < M) = 1$ for all n and $X_n \to X$ in probability, then $X_n \to X$ in rth moment for all $r > 0$.

A typical tool of proving almost sure convergence is **Borel-Cantelli Lemma**.

Lemma 2.1 Borel-Cantelli Lemma: *If $\{A_n, n \geq 1\}$ is a sequence of events for which $\sum_{i=1}^{\infty} P(A_n) < \infty$, then*

$$P(A_n \text{ occur infinitely often}) = 0.$$

The event $\{A_n$ occur infinitely often$\}$ contains all sample points which is a member of infinite number of A_n's. We will use i.o. for *infinitely often*. The fact that $X_n \to X$ almost surely is equivalent to

$$P(|X_n - X| \geq \varepsilon, \text{i.o. }) = 0$$

for all $\varepsilon > 0$. In view of Borel-Cantelli Lemma, if

$$\sum_{n=1}^{\infty} P(|X_n - X| \geq \varepsilon) < \infty,$$

for all $\varepsilon > 0$, then $X_n \to X$ almost surely.

Let $X_1, X_2, \ldots, X_n, \ldots$ be a sequence of independent and identically distributed (iid) random variables such that their second moment exists. Let $\mu = E(X_1)$ and $\sigma^2 = Var(X_1)$. Let $\bar{X}_n = n^{-1} \sum_{i=1}^{n} X_i$ so that $\{\bar{X}_n\}$ is a sequence of random variables too. By Chebyshev's inequality,

$$P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2}$$

for any give $\varepsilon > 0$. As $n \to \infty$, the probability converges to 0. Hence, we have shown $\bar{X}_n \to \mu$ in probability. Note that we may view μ as a random variable with a degenerate distribution.

The proof can be used to establish the almost sure convergence if the 4th moment of X_1 exists. In fact, the existence of the first moment is sufficient to establish the almost sure convergence of the sample mean of the i.i.d. random variables. The elementary proof under the first moment assumption only is long and complex. We present the followings without proofs.

Theorem 2.1 Law of Large Numbers*: Let $X_1, X_2, \ldots, X_n, \ldots$ be a sequence of independent and identically distributed (i.i.d.) random variables.*

(a) If $nP(|X_1| > n) \to 0$, then

$$\bar{X}_n - c_n \to 0$$

in probability, where $c_n = E\{X_1 I(|X_1| \le n)\}$.

(b) If $E|X_1| < \infty$, then

$$\bar{X}_n - E(X_1) \to 0$$

almost surely.

The existence of the first moment of a random variable is closely related to how fast $P(|X| > n)$ goes to zero as $n \to \infty$. Here we give an interesting inequality and a related result.

Let X be a positive random variable with finite expectation. That is, assume $P(X \ge 0) = 1$ and $E\{X\} < \infty$. Then we have

$$E\{X\} = \sum_{n=0}^{\infty} E\{XI(n < X \le n+1)\}.$$

Since

$$nI(n < X \le n+1) < XI(n < X \le n+1) < (n+1)I(n < X \le n+1)$$

for all $n = 0, 1, \ldots$. We get

$$\sum_{n=0}^{\infty} \{nP(n < X \le n+1)\} \le E\{X\} \le \sum_{n=0}^{\infty} \{(n+1)P(n < X \le n+1)\}.$$

Let $q_n = P(X > n)$ so that $P(n < X \leq n+1) = q_n - q_{n+1}$. We then have

$$\sum_{n=0}^{\infty} \{nP(n < X \leq n+1)\} = \sum_{n=0}^{\infty} q_{n+1}.$$

Consequently, if $E\{X\} < \infty$, then

$$\sum_{n=0}^{\infty} q_{n+1} = \sum_{n=0}^{\infty} P(X > n+1) < \infty.$$

If $X_1, X_2, \ldots, X_n, \ldots$ is a sequence of random variables with the same distribution as X, then we have

$$\sum_{n=0}^{\infty} P(X_n > n+1) < \infty.$$

By Borel-Cantelli Lemma, $X_n < n+1$ almost surely.

2.2 Uniform Strong law of large numbers

In many statistical problems, we must work on i.i.d. random variables indexed by some parameters. For each given parameter value, the (strong) law of large number is applicable. However, we are often interested in large sample properties of a parameter derived from the sum of such random variables. These properties can often be obtained based on the uniform convergence with probability one of these functions. Rubin (1956) gives a sufficient condition for such uniform convergence which is particularly simple to use.

Let $X_1, X_2, \ldots, X_n \ldots$ be a sequence of i.i.d. random variables taking values in an arbitrary space $\mathscr{X}$. Let $g(x;\theta)$ be a measurable function in x for each $\theta \in \Theta$. Suppose further that Θ is a compact parameter space.

Theorem 2.2 *Suppose there exists a function $H(\cdot)$ such that $E\{H(X)\} < \infty$ and that $|g(x;\theta)| \leq H(x)$ for all $\theta \in \Theta$. The parameter space Θ is compact. In addition, there exists A_j, $j = 1, 2, \ldots$ such that*

$$P(X_i \in \cup_{j=1}^{\infty} A_j) = 1$$

and $g(x;\theta)$ is continuous in θ uniformly on $x \in A_j$ for each j. Then, almost surely, and uniformly in $\theta \in \Theta$,

$$n^{-1}\sum_{i=1}^{n} g(X_i;\theta) \to E\{g(X_1;\theta)\}$$

and that $E\{g(X_1;\theta)\}$ is a continuous function in θ.

Proof: We may define $B_k = \cup_{j=1}^{k} A_j$ for $k = 1,2,\ldots$. Note that B_k is monotone increasing. The theorem condition implies that $P(X \in B_k) \to 1$ as $k \to \infty$ and therefore

$$H(X)\mathbb{1}(X \in B_k^c) \to 0$$

almost surely, where X is a random variable with the same distribution with X_1. By the dominant convergence theorem, the condition $E\{H(X)\} < \infty$ leads to

$$E\{H(X)\mathbb{1}(X \in B_k^c)\} \to 0$$

as $k \to \infty$. We now take note of

$$\begin{aligned}
&\sup_{\theta}\{\left|n^{-1}\sum_{i=1}^{n} g(X_i;\theta) - E\{g(X;\theta)\}\right|\} \\
&\quad\leq \sup_{\theta}\{\left|n^{-1}\sum_{i=1}^{n} g(X_i;\theta)\mathbb{1}(X_i \in B_k) - E\{g(X;\theta)\mathbb{1}(X_i \in B_k)\}\right|\} \\
&\qquad + \sup_{\theta}\{\left|n^{-1}\sum_{i=1}^{n} g(X_i;\theta)\mathbb{1}(X_i \notin B_k) - E\{g(X;\theta)\mathbb{1}(X_i \notin B_k)\}\right|\}.
\end{aligned}$$

The second term is bounded by

$$n^{-1}\sum_{i=1}^{n} H(X_i)\mathbb{1}(X_i \notin B_k) + E\{H(X)\mathbb{1}(X \in B_k^c)\} \to 2E\{H(X)\mathbb{1}(X \in B_k^c)\}$$

which is arbitrarily small almost surely. Because $H(X)$ dominants $g(X;\theta)$, these results show that the proof of the theorem can be carried out as as if $\mathscr{X} = B_k$ for some large enough k.

In other words, we need only prove this theorem when $g(x;\theta)$ is simply equicontinuous over x. Under this condition, for any $\varepsilon > 0$, there exist a finite number of θ values, $\theta_1, \theta_2, \ldots, \theta_m$ such that

$$\sup_{\theta\in\Theta} \min_{j} |g(x;\theta) - g(x;\theta_j)| < \varepsilon/2.$$

This also implies

$$\sup_{\theta\in\Theta}\min_{j}|E\{g(X;\theta)\}-E\{g(X;\theta_j)\}|<\varepsilon/2.$$

Next, we easily observe that

$$\sup_{\theta}\{\left|n^{-1}\sum_{i=1}^{n}g(X_i;\theta)-E\{g(X;\theta)\}\right|\}\leq\max_{1\leq j\leq m}\{\left|n^{-1}\sum_{i=1}^{n}g(X_i;\theta_j)-E\{g(X;\theta_j)\}\right|\}+\varepsilon.$$

The first term goes to 0 almost surely by the conventional strong law of large numbers and ε is an arbitrarily small positive number. This conclusion is therefore true.

2.3 Convergence in distribution

The concept of convergence in distribution is different from the modes of convergence given in the last section.

Definition 2.4 Convergence in distribution: *Let $X_1, X_2, \ldots, X_n, \ldots$ be a sequence of random variables, and X be another random variable. If*

$$P(X_n\leq x_0)\to P(X\leq x_0)$$

for all x_0 such that $F(x)=P(X\leq x)$ is continuous at $x=x_0$, then we say that $X_n\to X$ in distribution. We may also denote it as $X_n\xrightarrow{d}X$.

The convergence in distribution is not dependent on the probability space. Thus, we may instead discuss a sequence of distribution functions $F_n(x)$ and $F(x)$. If $F_n(x)\to F(x)$ at all continuous point of $F(x)$, then F_n converges to $F(x)$ in distribution. We sometimes mix up the random variables and their distribution functions. When we state that X_n converges to $F(x)$, the claim is the same as the distribution of X_n converges to $F(x)$.

It could happen that $F_n(x)$ converges at each x, but the limit, say $F(x)$, does not have the properties such as $\lim_{x\to\infty}F(x)=1$. In this case, $F_n(x)$ does not converge in distribution although the function sequence converges.

Example 2.3 *Let X be a positive random variable and $X_n = nX$ for $n = 1, 2, \ldots$. It is seen that $P(X_n < x) \to 0$ for any finite x. Let $F_n(x)$ denote the distribution function of X_n, we have $F_n(x) \to 0$. However, this convergence is not in the mode of "in distribution".*

When $F_n \to F$ in distribution, there may not be any corresponding random variables under discussion. It is always possible to construct a probability space and a sequence of random variables $\{X_n\}_{n=1}^{\infty}$ and X such that their distribution functions are the same as F_n and F. Furthermore, the construction can be done such that $X_n \to X$ almost surely.

Theorem 2.3 Skorohod representation Theorem. *Suppose $F_n \to F$ in distribution. There exists a probability space and a sequence of random variables $\{X_n\}_{n=1}^{\infty}$ and X such that $X_n \to X$ almost surely.*

Using this result, one may show that if $X_n \to X$ in distribution and $g(x)$ is a continuous function, then $g(X_n) \to g(X)$ in distribution.

We end this section by presenting two useful results. (1) $X_n \to X$ in distribution if and only if $E[g(X_n)] \to E[g(X)]$ for all bounded and continuous function g. (2) When $X_1, X_2, \ldots$ are random vectors of finite dimension, then $X_n \to X$ in distribution if and only if for any non-random vector a, $a^\tau X_n \to a^\tau X$ in distribution.

Example 2.4 *Let $X_1, X_2, \ldots, X_n$ be a sequence of iid exponentially distributed random variables. Their common cumulative distribution family is given by*

$$F(x) = 1 - \exp(-x)$$

for $x \geq 0$.

Let $X_{(n)} = \max\{X_1, \ldots, X_n\}$ and $X_{(1)} = \min\{X_1, \ldots, X_n\}$.

It is seen that

$$P(nX_{(1)} > x) = \{\exp(-x/n)\}^n = \exp(-x).$$

Hence, $nX_{(1)} \to X_1$ in distribution.

On the other hand, we find

$$P\{X_{(n)} - \log n < x\} = \{1 - n^{-1}\exp(-x)\}^n \to \exp(-e^{-x}).$$

The right hand side is a cumulative distribution function. Hence, $X_{(n)} - \log n$ converges in distribution to a distribution with cumulative distribution function $\exp(-e^{-x})$. We call it type I extreme value distribution.

2.4 Central limit theorem

The most important example of the convergence in distribution is the classical central limit theorem. It presents an important case when a commonly used statistic is asymptotically normal. The simplest version is as follows. By $N(\mu, \sigma^2)$, we mean the normal distribution with mean μ and variance σ^2.

Theorem 2.4 Classical Central Limit Theorem: *Let $X_1, X_2, \ldots$ be a sequence of iid random variables. Assume that both $\mu = E(X_1)$ and $\sigma^2 = Var(X_1)$ exist. Then, as $n \to \infty$,*

$$\sqrt{n}[\bar{X}_n - \mu] \to N(0, \sigma^2)$$

in distribution, where $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$.

It may appear illogical to some that we start with a sequence of random variables, but end up with a normal distribution. As already commented in the last section, we interpret both sides as their corresponding distribution functions.

If X_n's do not have the same distribution, then having common mean and variance is not sufficient for the asymptotic normality of the sample mean. A set of nearly necessary and sufficient conditions is the Lindberg condition. For most applications, we recommend the verification of the Liapounov condition.

Theorem 2.5 Central Limit Theorem under Liapounov Condition: *Let $X_1, X_2, \ldots$ be a sequence of independent random variables. Assume that both $\mu_i = E(X_i)$ and $\sigma_i^2 = Var(X_i)$ exist. Further, assume that for some $\delta > 0$,*

$$\frac{\sum_{i=1}^n E|X_i - \mu_i|^{2+\delta}}{[\sum_{i=1}^n \sigma_i^2]^{1+\delta/2}} \to 0$$

as $n \to \infty$. Then, as $n \to \infty$,

$$\frac{\sum_{i=1}^n (X_i - \mu_i)}{\sqrt{\sum_{i=1}^n \sigma_i^2}} \to N(0,1)$$

in distribution.

The central limit theorem for random vectors is established through examining the convergence of $a^\tau X_n$ for all possible non-random vector a.

2.5 Big and small o, Slutsky's theorem

There are many important statistics that are not straight sum of independent random variables. At the same time, many are also asymptotically normal. Many of such results are proved with the help of Slutsky's theorem and with the concepts of big and small o's.

Let a_n be a sequence of positive numbers and X_n be a sequence of random variables. If

$$X_n/a_n \to 0$$

in probability, we say $X_n = o_p(a_n)$. In general, the definition is meaningful only if a_n is a monotone sequence. If instead, for any given $\varepsilon > 0$, there exist positive constants M and N such that whenever $n > N$,

$$P(|X_n/a_n| < M) \geq 1 - \varepsilon,$$

then we say that $X_n = O_p(a_n)$. In most textbooks, the positiveness of a_n is not required. Not requiring positiveness does not change the essence of the current definition. Sticking to positiveness is helpful at avoiding some un-intended abuse of these concepts.

We love to compare statistics under investigation to $n^{1/2}, n, n^{-1/2}$ and so on. If $X_n = o_p(n^{-1})$, it implies that X_n converges to 0 faster than the rate of n^{-1}. If $X_n = O_p(n^{-1})$, it implies that X_n converges to 0 no slower than the rate of n^{-1}. Most importantly, when $X_n = O_p(n)$, it does not imply X_n has a size of n when n is large. Even if $X_n = 0$ for all n, it is still true that $X_n = O_p(n)$.

Example 2.5 *If $E|X_n| = o(1)$, then $X_n = o_p(1)$.*

Proof: By Markov inequality, for any $M > 0$, we have

$$P(|X_n| > M) \leq E|X_n|/M = o(1).$$

Hence, $X_n = o_p(1)$.

The reverse of the above example is clearly wrong.

Example 2.6 *Suppose $P(X_n = 0) = 1 - n^{-1}$ and $P(X_n = n) = n^{-1}$. Then $X_n = o_p(n^{-m})$ for any fixed $m > 0$. Yet we do not have $E\{X_n\} = o(1)$.*

While the above example appears in almost all textbooks, it is not unusual to find such misconception appears in research papers in some disguised forms.

Example 2.7 *If $X_n = O_p(a_n)$ and $Y_n = O_p(b_n)$ for two* **positive** *sequences of real numbers a_n and b_n, then*

(i) $X_n + Y_n = O_p(a_n + b_n)$;

(ii) $X_n Y_n = O_p(a_n b_n)$.

However, $X_n - Y_n = O_p(a_n - b_n)$ or $X_n/Y_n = O_p(a_n/b_n)$ are not necessarily true.

Example 2.8 *Suppose $X_1, \ldots, X_n$ is a set of iid random variables from Poisson distribution with mean θ. Let $\bar{X}_n$ be the sample mean.*

Then, we have

(1) $\exp(\bar{X}_n) = \exp(\theta) + o_p(1)$.

(2) $\exp(\bar{X}_n) = \exp(\theta) + O_p(n^{-1/2})$.

Let us first present a simplified version of Slutsky's Theorem.

Theorem 2.6 *Suppose $X_n \to X$ in distribution, and $Y_n = o_p(1)$, then $X_n + Y_n \to X$ in distribution.*

PROOF: Let $F_n(x)$ and $F(x)$ be the cumulative distribution functions of X_n and X. Let x be a continuous point of $F(x)$. For any given $\varepsilon > 0$, according to some real analysis result, we can always find $0 < \varepsilon' < \varepsilon$ such that $F(x)$ is also continuous at $x + \varepsilon'$.

Since $Y_n = o_p(1)$, we have for any $\delta > 0$ and $\varepsilon > 0$, there exists an N such that when $n > N$,

$$P(|Y_n| \le \varepsilon) > 1 - \delta.$$

Let ε be chosen such that $x + \varepsilon$ is a continuous point of F. Hence, when $n > N$, we have

$$P(X_n + Y_n \le x) \le P(X_n \le x + \varepsilon) + \delta \to F(x + \varepsilon) + \delta.$$

Since δ can be arbitrarily small, we have shown

$$\limsup P(X_n + Y_n \le x) \le F(x + \varepsilon)$$

for all ε such that $x+\varepsilon$ is a continuous point of F. As indicated earlier, such ε can also be chosen arbitrarily small, we may let $\varepsilon \to 0$. Consequently, by the continuity of F at x, we have

$$\limsup P(X_n + Y_n \le x) \le F(x).$$

Similarly, we can show that

$$\liminf P(X_n + Y_n \le x) \ge F(x).$$

Two inequalities together imply $X_n + Y_n \to X$ in distribution. $\diamondsuit$

If $F(x)$ is a continuous function, then we can save a lot of trouble in the above proof.

The simplified Slutsky's Theorem I presented above is also refereed as delta-method when it is used as a tool for proving asymptotic results. In a nut shell, it simply states that adding a $o_p(1)$ quantity to a sequence of random variables does not change the limiting distribution.

2.6 Asymptotic normality for functions of random variables

Suppose we already know that for some $a_n \to \infty$, $a_n(Y_n - b_n) \xrightarrow{d} Y$. What do we know about the distribution of $g(Y_n) - g(b_n)$? The first observation is, if b_n does not have a limit, then even if g is a smooth function, the difference is still far from determined. In general, $g(Y_n) - g(b_n)$ depends on the slope of g near b_n. Hence we only consider the case where b_n is a constant that does not depend on n.

Theorem 2.7 *Assume that $a_n(Y_n - \mu) \to Y$ in distribution, $a_n \to \infty$, and $g(\cdot)$ is continuously differentiable in a neighborhood of μ. Then*

$$a_n[g(Y_n) - g(\mu)] \to g'(\mu)Y$$

in distribution.

Proof: Using the mean value theorem

$$a_n[g(Y_n) - g(\mu)] = g'(\xi_n)[a_n(Y_n - \mu)],$$

for some value of ξ_n between Y_n and μ. Since $a_n \to \infty$, we must have $Y_n \to \mu$ in probability. Hence we also have $\xi_n \xrightarrow{p} \mu$. Consequently, the differentiability of g at μ implies

$$g'(\xi_n) - g(\mu) = o_p(1)$$

and

$$a_n[g(Y_n) - g(\mu)] = g'(\mu)[a_n(Y_n - \mu)] + o_p(1).$$

The result follows the Slutsky's theorem. ♢

The result and the proof are presented for the case when X_n and Y are one-dimensional. It can be easily generalized to vector cases. When a_n does not converge to any constant, out idea should still apply. It is not smart to declare that the asymptotic does not work because the conditions of Theorem 2.7 are not satisfied.

2.7 Sum of random number of random variables

Sometimes we need to work with the sum of random number of random variables. One such example is the total amount of insurance claims in a month.

Theorem 2.8 *Let $\{X_i, i = 1,2,\ldots\}$ be i.i.d. random variables with mean μ and variance σ^2. Let $\{N_i, i = 1,2,\ldots\}$ be a sequence of integer valued random variables which is independent of $\{X_i, i = 1,2,\ldots\}$, and $P(N_n > M) \to 1$ for any M as $n \to \infty$. Then*

$$N_n^{-1/2} \sum_{j=1}^{N_n} (X_j - \mu) \to N(0, \sigma^2)$$

in distribution.

Proof: For simplicity, assume $\mu = 0$, $\sigma^2 = 1$ and let $Y_n = n^{-1/2} \sum_{i=1}^n X_i$. The classical central limit theorem implies that for any real value x and a positive constant ε, there exists a constant M, such that whenever $n > M$,

$$|P(Y_n \le x) - \Phi(x)| \le \varepsilon.$$

From the independence assumption,

$$P(Y_{N_n} \le x) = \sum_{m=1}^{\infty} P(Y_m \le x) P(N_n = m).$$

Hence,

$$\begin{aligned} |P(Y_{N_n} \le x) - \Phi(x)| &= |\sum_{m=1}^{\infty} \{P(Y_m \le x) - \Phi(x)\} P(N_n = m)| \\ &\le |\sum_{m \ge M} P(Y_m \le x) P(N_n = m) - \Phi(x)| + P(N_n < M) \\ &\le \varepsilon + P(N_n < M) \to \varepsilon. \end{aligned}$$

From the arbitrariness of the choice of ε, we conclude that $P(Y_{N_n} \le x) \to \Phi(x)$ for all x. Hence, the theorem is proved. $\diamondsuit$

2.8 Assignment problems

1. Prove that the set $\{\omega : \lim X_n(\omega) \text{ exists}\}$ is measurable.

2. Identify an almost surely convergence subsequence in the context of Example 2.1.

3. Prove Borel-Cantelli Lemma.

4. Suppose that there exists a nonrandom constant M such that $P(|X_n| < M) = 1$ for all n and $X_n \to X$ in probability. Show that $X_n \to X$ in rth moment for all $r > 0$.

5. Show that if $X_n \to X$ almost surely, then $X_n \to X$ in probability.

6. Using Borel-Cantelli Lemma to show that the sample mean $\bar{X}_n$ of an i.i.d. sample converges to its mean almost surely if $E|X_1|^4 < \infty$.

7. Show that if $X_n \to X$ in distribution and $g(x)$ is a continuous function, then $g(X_n) \to g(X)$ in distribution.

   Furthermore, give an example of non-continuous $g(x)$ such that $g(X_n)$ does not converge to $g(X)$ in distribution.

8. Prove that $X_n \to X$ in distribution if and only if $E[g(X_n)] \to E[g(X)]$ for all bounded and continuous function g.

9. Let $X_1, \ldots, X_n$ be an i.i.d. sample from uniform distribution on [0, 1]. Find the limiting distribution of $nX_{(1)}$, where $X_{(1)} = \min\{X_1, \ldots, X_n\}$ when $n \to \infty$.

10. Let $X_1, \ldots, X_n$ be an i.i.d. sample from standard normal distribution. Find a non-degenerating limiting distribution of $a_n(X_{(1)} - b_n)$ with appropriate choices of a_n and b_n.

11. Suppose F_n and F are a sequence of one-dimensional cumulative distribution functions and that $F_n \xrightarrow{d} F$. Show that

$$\sup_x |F_n(x) - F(x)| \to 0$$

as $n \to \infty$ if $F(x)$ is a continuous function.

Give a counter–example when F is not continuous.

12. Suppose F_n and F are a sequence of absolutely continuous one-dimensional cumulative distribution functions and that $F_n \xrightarrow{d} F$. Let $f_n(x)$ and $f(x)$ be their density functions. Give a counter example to

$$\int |f_n(x) - f(x)|dx \to 0$$

as $n \to \infty$.

Prove the the above limiting conclusion is true when $f_n(x) \to f(x)$ at all x.

Are there any similar results for discrete distributions?

13. Suppose that $\{X_{ni}, i = 1, \ldots, n\}_{n=1}^{\infty}$ is a sequence of sets of random variables. It is known that

$$\max P(|X_{ni}| > n^{-2}) \to 0$$

as $n \to \infty$. Does it imply that $\sum_{i=1}^n X_{ni} = o_p(n^{-1})$? What is the order of $\max_{1 \le i \le n}\{X_{ni}\}$?

14. Suppose that $X_n = O_p(n^2)$ and $Y_n = o_p(n^2)$. Is it true that $Y_n/X_n = o_p(1)$?

15. Suppose that $X_n = O_p(a_n)$ and $Y_n = O_p(b_n)$. Prove that $X_nY_n = O_p(a_nb_n)$.

16. Suppose that $X_n = O_p(a_n)$ and $Y_n = O_p(b_n)$. Give a counter example to $X_n - Y_n = O_p(a_n - b_n)$.

17. Suppose we have a sequence of random variable $\{X_n\}$ such that $X_n \xrightarrow{d} X$. Show that $X_n = O_p(1)$.

18. Suppose $X_n \xrightarrow{d} X$ and $Y_n = 1 + o_p(1)$. Is it true that $X_n/Y_n \xrightarrow{d} X$?

19. Let $\{X_i\}_{i=1}^{\infty}$ be a sequence of i.i.d. random variables. Show that $X_n = O_p(1)$. Is it true that $\sum_{i=1}^{n} X_i = O_p(n)$?

20. Assume that $a_n(Y_n - \mu) \to Y$ in distribution, $a_n \to \infty$, and $g(\cdot)$ is continuously differentiable in a neighborhood of μ.

    Suppose $g'(\mu) = 0$ and $g''(x)$ is continuous and non-zero at $x = \mu$. Obtain a limiting distribution of $g(Y_n) - g(\mu)$ under an appropriate scale.

Chapter 3

Empirical distributions, moments and quantiles

Let $X, X_1, X_2, \ldots$ be i.i.d. random variables. Let $m_k = \mathbb{E}\{X^k\}$ and $\mu_k = \mathbb{E}\{(X - m_1)^k\}$ for $k = 1, 2, \ldots$. We may also use notation μ for m_1, and σ^2 for μ_2. We call m_k the kth moment and μ_k the kth central moment.

With n i.i.d. observations of X, a corresponding empirical distribution function F_n is constructed by placing at each observation X_i a mass n^{-1}. That is,

$$F_n(x) = \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}(X_i \leq x), \quad -\infty < x < \infty.$$

We also call $F_n(x)$ the empirical distribution. It is a natural estimator of the cumulative distribution function of X.

Assume that $F_n(x)$ is not random for the moment. Then if a random variable X^* has cumulative distribution $F_n(x)$, we would find its moments are given by

$$m_k^* = \frac{1}{n} \sum_{i=1}^{n} X_i^k \text{ and } \mu_k^* = \frac{1}{n} \sum_{i=1}^{n} (X_i - m_1^*)^k.$$

We denote them as $\hat{m}_k$ and $\hat{\mu}_k$ because they are natural estimates of m_k and μ_k. We use $\bar{X}_n$ for the sample mean and S_n^2 for the sample variance.

Since we have reasons to believe that F_n is a good estimator of F, it may also be true that $\psi(F_n)$ will estimate $\psi(F)$ for any reasonable functional ψ. In this chapter, we discuss large sample properties of $\psi(F_n)$.

3.1 Properties of sample moments

Moments of a distribution family are very important parameters. Sample moments provide natural estimates. Many other parameters are functions of moments, therefore, estimates can be obtained by using the functions of sample moments. This is the so-called method of moments.

If the relevant moments of X_i exist, we can easily show

1. $\hat{m}_k \xrightarrow{a.s.} m_k$.
2. $n^{1/2}[\hat{m}_k - m_k] \xrightarrow{d} N(0, m_{2k} - m_k^2)$.
3. $\mathbb{E}(\hat{m}_k) = m_k$; $n\text{VAR}(\hat{m}_k) = m_{2k} - m_k^2$.

Before we work on the central sample moments $\hat{\mu}_k$, let us first define

$$b_k = \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^k, \quad k = 1, 2, \ldots.$$

If we replace X_i by $X_i - \mu$ in $\hat{m}_k$, it becomes b_k. Obviously, $b_k \to \mu_k$ almost surely for all k when the kth moment of X is finite.

Theorem 3.1 *Let μ_k, $\hat{\mu}_k$ and so on are defined the same way as above. Assume that the kth moment of X is finite. Then, we have*

(a) $\hat{\mu}_k \xrightarrow{a.s.} \mu_k$ *almost surely.*

(b) $\mathbb{E}\hat{\mu}_k - \mu_k = \{\frac{1}{2}k(k-1)\mu_{k-2}\mu_2 - k\mu_k\}n^{-1} + O(n^{-2})$.

(b) $\sqrt{n}\{\hat{\mu}_k - \mu_k\} \xrightarrow{d} N(0, \sigma_k^2)$ *when we also have $\mathbb{E}\{X^{2k}\} < \infty$, where*

$$\sigma_k^2 = \mu_{2k} - \mu_k^2 - 2k\mu_{k-1}\mu_{k+1} + k^2\mu_2\mu_{k-1}^2.$$

PROOF The proof of conclusion (a) is straightforward.

(b). Without loss of generality, let us assume $\mu = 0$ to make the presentation simpler. It is seen that

$$\begin{aligned}
\hat{\mu}_k &= n^{-1}\sum_{i=1}^{n}(X_i-\bar{X})^k \\
&= n^{-1}\sum_{i=1}^{n}\left\{\sum_{j=0}^{k}\binom{k}{j}(-1)^j X_i^{k-j}\bar{X}^j\right\} \\
&= n^{-1}\sum_{i=1}^{n}X_i^k + n^{-1}\sum_{i=1}^{n}\left\{\sum_{j=1}^{k}\binom{k}{j}(-1)^j X_i^{k-j}\bar{X}^j\right\} \\
&= b_k + b_1\sum_{j=1}^{k}\binom{k}{j}(-1)^j b_{k-j}b_1^{j-1}.
\end{aligned}$$

This is the the first equality in (b).

For the second equality, note that $\mathbb{E}\{b_k\} = \mu_k$. Thus, we get

$$\mathbb{E}\{\hat{\mu}_k\} - \mu_k = \sum_{j=1}^{k}\binom{k}{j}(-1)^j\mathbb{E}\{b_1^j b_{k-j}\}.$$

We next study the order of these expectations term, $\mathbb{E}\{b_1^{j-1}b_{k-j}\}$ for $j = 1,2,\ldots,k$, term by term.

When $j = 1$, we have

$$\mathbb{E}\{b_1 b_{k-1}\} = n^{-2}\sum_{i,l}\mathbb{E}\{X_i X_l^{k-1}\}.$$

Due to independence and that X_i's have mean 0, the summand is zero unless $i = l$ and there are only n of them. From $\mathbb{E}X_i^k = \mu_k$, we get

$$\mathbb{E}\{b_1 b_{k-1}\} = n^{-1}\mu_k.$$

When $j = 2$, we have

$$\mathbb{E}\{b_1^2 b_{k-2}\} = n^{-3}\sum_{i,l,m}\mathbb{E}\{X_i X_l X_m^{k-2}\}.$$

A term in the above summation has nonzero expectation only if $i = l$. When $i = l$, we have two cases where $i = l = m$ and $i = j \neq m$. They have n and $n(n-1)$ terms

in the summation respectively. The corresponding expectations are given by μ_k and $\mu_2\mu_{k-2}$. Hence, we get

$$\mathbb{E}\{b_1^2 b_{k-2}\} = n^{-1}(\mu_k + \mu_2\mu_{k-2}) + O(n^{-2}).$$

When $j \geq 3$, we have

$$\mathbb{E}\{b_1^j b_{k-j}\} = n^{-(j+1)} \sum_{i_1,i_2,\ldots,i_j,l} \mathbb{E}\{X_{i_1}X_{i_2}\cdots X_{i_j}X_l^{k-j}\}.$$

The individual expectations are non-zero only if $i_1, i_2, \ldots, i_j$ are paired up with another index including possibly also l. Hence, for terms with nonzero expectation, there are at most $j-2$ different indices in $\{i_1, i_2, \ldots, i_j, l\}$. The total number of such indices is no more than n^{j-1}. Since $\mathbb{E}\{X_{i_1}X_{i_2}\cdots X_{i_j}X_l^{k-j}\} < \infty$, we must have

$$\mathbb{E}\{b_1^j b_{k-j}\} = O(n^{-2}).$$

Combining the calculations for $j = 1,2$ and for $j \geq 3$, we get the conclusion.

(c). We seek to use the Slutzky's theorem in this proof. This amounts to expand the random quantity into a leading term whose limiting distribution can be shown by a classical result, and an $o_p(1)$ which does not alter the outcome of the limiting distribution.

Since $b_1 = O_p(n^{-1/2})$, $b_1^2 = O_p(n^{-1})$, and $b_{k-j} = O_p(1)$, we get $b_1^j b_{k-j} = O(n^{-1})$ for $j \geq 2$. Consequently, we find

$$\begin{aligned}\sqrt{n}\{\hat{\mu}_k - \mu_k\} &= \sqrt{n}\{b_k - \mu_k - kb_1\mu_{k-1}\} + \sqrt{n}kb_1(b_{k-1} - \mu_{k-1}) + O_p(n^{-1/2}) \\ &= \sqrt{n}\{b_k - \mu_k - kb_1\mu_{k-1}\} + O_p(n^{-1/2}).\end{aligned}$$

The last equality is a resultant of $b_1(b_{k-1} - \mu_{k-1}) = O_p(n^{-1})$. It is seen that

$$b_k - \mu_k - kb_1\mu_{k-1} = n^{-1}\sum_{i=1}^{n}\{X_i^k - \mu_{k-1}X_i - \mu_k\}$$

which the sum of i.i.d. random variables. It is trivial to verify that $\mathbb{E}\{X_i^k - \mu_{k-1}X_i - \mu_k\} = 0$ and $\text{VAR}\{X_i^k - \mu_{k-1}X_i - \mu_k\} = \mu_{2k} - \mu_k^2 - 2k\mu_{k-1}\mu_{k+1} + k^2\mu_2\mu_{k-1}^2$. Applying the classical central limit theorem to $n^{-1/2}\sum_{i=1}^{n}\{X_i^k - \mu_{k-1}X_i - \mu_k\}$, we get the conclusion. $\diamondsuit$

The same technique can be used to show that $\mathbb{E}(\bar{X}_n - \mu)^k = O(n^{-k/2})$ when k is a positive even integer; and that $\mathbb{E}(\bar{X}_n - \mu)^k = O(n^{-(k+1)/2})$ when k is a positive odd integer. The second result is, however, not as obvious.

Here is a proof. The claim is the same as $\mathbb{E}(\sum_{i=1}^n X_i)^k = O(n^{k/2})$ or $O(n^{(k+1)/2})$ when k is odd. In the expansion of this summation, all terms have form

$$X_{i_1}^{j_1} \cdots X_{i_m}^{j_m}$$

with $j_1, \ldots, j_m > 0$ and $j_1 + \cdots + j_m = k$. Its expectation equals 0 whenever one of them equals 1. Thus, the size of m is at most $k/2$ or $(k-1)/2$ when k is odd. Since, each $i_1, \ldots, i_m$ has at most n choices, the total number of such terms is no more than $n^{k/2}$ of $O(n^{(k-1)/2})$ when k is odd. As their moments have an upper bound, the claim is proved.

Theorem 3.2 *Assume that the kth moment of X_1 exists and $\bar{X}_n = n^{-1}\sum_{i=1}^n X_i$. Then*

(a) $\mathbb{E}(\bar{X}_n - \mu)^k = O(n^{-k/2})$ *when k is a positive even integer and that* $\mathbb{E}(\bar{X}_n - \mu)^k = O(n^{-(k+1)/2})$.

(b) $\mathbb{E}|\bar{X}_n - \mu|^k = O(n^{-k/2})$ *when $k \geq 2$.*

PROOF:

(a) The claims are the same as $\mathbb{E}(\sum_{i=1}^n X_i)^k = O(n^{k/2})$ or $O(n^{(k+1)/2})$ when k is odd. We have a generic form of expansion

$$\Big(\sum_{i=1}^n X_i\Big)^k = \sum X_{i_1}^{j_1} \cdots X_{i_m}^{j_m}$$

such that the summation is over all combinations of $j_1, \ldots, j_m > 0$ and $j_1 + \cdots + j_m = k$.

The expectation of $X_{i_1}^{j_1} \cdots X_{i_m}^{j_m}$ equals 0 whenever one of $j_1, \ldots, j_m$ equals 1. Thus, the terms with nonzero expectation must have $m \leq k/2$, or $m \leq (k-1)/2$ when k is odd. Since, each $i_1, \ldots, i_m$ takes at most n values, the total number of nonzero expectation terms is no more than $n^{k/2}$ of $O(n^{(k-1)/2})$ when k is odd. Since their moments are bounded by a common constant, the claims must be true.

(b) The proof of this result becomes trivial based on the inequality in the next theorem. We omit the actual proof here.

Theorem 3.3 *Assume that Y_i, $i = 1,2,\ldots,n$ are independent random variables with $\mathbb{E}(Y_i) = 0$ for all i. Then, for some $k > 1$,*

$$A_k \mathbb{E}\{\sum_{i=1}^{n} Y_i^2\}^{k/2} \leq \mathbb{E}\{|\sum_{i=1}^{n} Y_i|^k\} \leq B_k \mathbb{E}\{\sum_{i=1}^{n} Y_i^2\}^{k/2}$$

where A_k and B_k are some positive constants not depending on n.

This inequality is attributed to Marcinkiewics-Zygmund and the proof can be found in Chow and Teicher (1978, 1st Edition, page 356). Its proof is somewhat involved.

3.2 Empirical distribution function

For each fixed x, $F_n(x)$ is the sample mean of $Y_i = I(X_i \leq x)$, $i = 1,2,\ldots,n$. Since Y_i's are i.i.d. random variables and they have finite moments to any order, the standard large sample results apply. We can easily claim:

1. $F_n(x) \xrightarrow{a.s.} F(x)$ for each fixed x, and in any order of moments.
2. $\sqrt{n}\{F_n(x) - F(x)\} \xrightarrow{d} N(0, \sigma^2(x))$ with $\sigma^2(x) = F(x)\{1 - F(x)\}$.
3. $F_n(x) - F(x) = O_p(n^{-1/2})$.

The conclusion 3 is a corollary of conclusion 2. A direct proof can be done by using Chebyshev's inequality:

$$P(\sqrt{n}|F_n(x) - F(x)| > M) \leq \frac{\sigma^2(x)}{M^2}$$

whose right hand side can be made arbitrarily small with a proper choice of M.

Recall that if $F(x)$ is continuous, then the convergence of $F_n(x)$ at every x implies the uniform convergence in x. That is, $D_n = \sup_x |F_n(x) - F(x)|$ converges to 0 almost surely. The statistic D_n is called the Kolmogorov-Smirnov distance and it is used for the goodness of fit test. In fact, when F is continuous and univariate, it is known that

$$P(D_n > d) \leq C \exp\{-2nd^2\}$$

for all n and d, and C is an absolute constant. If X is a random vector, this result remains true with 2 replaced by $2-\varepsilon$, and C then depends on the dimension and ε.

In addition, under same conditions,

$$\lim_{n\to\infty} P(n^{1/2}D_n \le d) = 1 - 2\sum_{j=1}^{\infty}(-1)^{j+1}\exp(2j^2d^2).$$

We refer to Serfling (1980) for more results.

3.3 Sample quantiles

Let $F(x)$ be a cumulative distribution function. We define, for any $0 < p < 1$, its pth quantile as

$$\xi_p = F^{-1}(p) = \inf\{x : F(x) \ge p\}.$$

Intuitively, if ξ_p is the pth quantile of $F(x)$, we should have $F(\xi_p) = p$. The above definition clearly does not guarantee its validity. The problem arises when $F(x)$ jumps at ξ_p. We can, however, prove the following:

Theorem 3.4 *Let F be a distribution function. The function $F^{-1}(t)$, $0 < t < 1$, is nondecreasing and left-continuous, and satisfies*

1. $F^{-1}(F(x)) \le x$, $-\infty < x < \infty$,
2. $F(F^{-1}(t)) \ge t$, $0 < t < 1$,
3. $F(x) \ge t$ *if and only if* $x \ge F^{-1}(t)$.

PROOF: We first show that the inverse is monotone. When $t_1 < t_2$, we have

$$\{x : F(x) \ge t_1\} \supset \{x : F(x) \ge t_2\}.$$

The lowest value in a smaller set is larger than the lowest value in a larger set. Hence,

$$\inf\{x : F(x) \ge t_1\} \le \inf\{x : F(x) \ge t_2\}.$$

which is $F^{-1}(t_1) \le F^{-1}(t_2)$ or monotonicity.

To prove the left continuity, let $\{t_k\}_{k=1}^{\infty}$ be an increasing sequence taking values between 0 and 1 with limit t_0. We hence have $F^{-1}(t_k)$ is an increasing sequence with upper bound $F^{-1}(t_0)$. Hence, it has a limit. We wish to show $F^{-1}(t_k) \to F^{-1}(t_0)$. If not, let $x \in (\lim F^{-1}(t_k), F^{-1}(t_0))$. This implies

$$t_0 > F(x) \geq t_k$$

for all k. This is not possible when $\lim t_k = t_0$.

1. By definition, for any y such that $F(y) \geq F(x)$, we have $y \geq F^{-1}(F(x))$. This remains to be true when $y = x$, hence $x \geq F^{-1}(F(x))$.

2. For any $y > F^{-1}(t)$, we have $F(y) \geq t$ by definition. Let $y \to F^{-1}(t)$ from right, and from the right-continuity, we must have $F(F^{-1}(t)) \geq t$.

3. This is the consequence of (1) and (2). $\diamondsuit$.

With an empirical distribution function $F_n(x)$, we define the empirical pth quantile $F_n^{-1}(p) = \hat{\xi}_p$. What properties does this estimator have?

In order for $\hat{\xi}_p$ to behave, some conditions on $F(x)$ seem necessary. For example, if $F(x)$ is a distribution which place 50% probability each at $+1$ and -1. The median of $F(x)$ equals -1 by our definition. The median of $F_n(x)$ equals -1 whenever less than 50% of observations are equal to -1 and it equals 1 otherwise. The median is not meaningful for this type of distributions. To be able to differentiate between ξ_p and $\xi_p \pm \delta$, it is most desirable that $F(x)$ strictly increase over this range.

Here is the consistency result for $\hat{\xi}_p$. Note that $\hat{\xi}_p$ depends on n although this fact is not explicit in its notation.

Theorem 3.5 *Let $0 < p < 1$. If ξ_p is the unique solution x of $F(x-) \leq p \leq F(x)$, then $\hat{\xi}_p \to \xi_p$ almost surely.*

Proof: For every $\varepsilon > 0$, by the uniqueness condition and the definition of ξ_p, we have

$$F(\xi_p - \varepsilon) < p < F(\xi_p + \varepsilon).$$

It has been shown earlier that $F_n(\xi_p \pm \varepsilon) \to F(\xi_p \pm \varepsilon)$ almost surely. This implies that

$$\xi_p - \varepsilon \le \hat{\xi}_p \le \xi_p + \varepsilon$$

almost surely. Since the size of ε is arbitrary, we must have $\hat{\xi}_p \to \xi_p$ almost surely. $\diamondsuit$.

If you like mathematics, the last sentence in the proof can be made more rigorous.

Theorem 3.6 *Let* $0 < p, < 1$. *If* F *is differentiable at* ξ_p *and* $F'(\xi_p) > 0$, *then*

$$\sqrt{n}F'(\xi_p)[\hat{\xi}_p - \xi_p] \xrightarrow{d} N(0, p(1-p)).$$

Proof: For any real number x, we have

$$(\sqrt{n}(\hat{\xi}_p - \xi_p) \le x\ ) = (\hat{\xi}_p \le \xi_p + \frac{x}{\sqrt{n}}).$$

By definition of the sample quantile, the above event is the same as the following event:

$$F_n(\xi_p + \frac{x}{\sqrt{n}}) \ge p.$$

Because F has positive derivative at ξ_p, we have $F(\xi_p) = p$. Thus,

$$\begin{aligned} & P\Big(\sqrt{n}[\hat{\xi}_p - \xi_p] \le x\Big) \\ &= P\Big(F_n(\xi_p + \frac{x}{\sqrt{n}}) - F(\xi_p + \frac{x}{\sqrt{n}}) \ge F(\xi_p) - F(\xi_p + \frac{x}{\sqrt{n}})\Big) \\ &= P\Big(F_n(\xi_p + \frac{x}{\sqrt{n}}) - F(\xi_p + \frac{x}{\sqrt{n}}) \ge -\frac{x}{\sqrt{n}}F'(\xi_p) + o(\frac{1}{\sqrt{n}})\Big) \\ &= P\Big(\sqrt{n}[F_n(\xi_p + \frac{x}{\sqrt{n}}) - F(\xi_p + \frac{x}{\sqrt{n}})] \ge -xF'(\xi_p) + o(1)\Big). \end{aligned}$$

By Slutsky's theorem, for the sake of deriving the limiting distribution, the term $o(1)$ can be ignored if the resulting probability has a limit. The resulting probability has limit as the c.d.f. of $N(0, [F'(\xi_p)]^2 p(1-p))$ by applying the central limit theorem for double arrays. $\diamondsuit$.

If $F(x)$ is absolutely continuous, then $F'(\xi_p) = f(\xi_p)$ the density function. To be more specific, let $p = 0.5$ and hence $\xi_{0.5}$ is the median. Thus, the efficiency of the sample median depends on the size of the density at the median.

If $F(x)$ is the standard normal, then $f(\xi_{0.5}) = \frac{1}{\sqrt{2\pi}}$. The asymptotic variance is hence $0.5^2 * (2\pi) = \frac{\pi}{2}$. In comparison, the sample mean has asymptotic variance 1 which is smaller. Both mean and median are the same location parameter for normal distribution family. Therefore, the sample mean is a more efficient estimator for the location parameter than the sample median. If, however, the distribution under consideration is double exponential, then the value of the density function at median is 0.5. Hence the asymptotic variance of the sample median is 1. At the same time, the sample mean has asymptotic variance 2. Thus, in this case, the sample median is more efficient.

If we take the more extreme example when $F(x)$ has Cauchy distribution, then the sample mean has infinite variance. The sample median is far superior. For those who advocate robust estimation, they point out that not only the sample median is robust, but also it can be more efficient when the model deviates from normality.

3.4 Inequalities on bounded random variables

We often work with bounded random variables. There are many particularly sharp inequalities for the sum of bounded random variables.

Theorem 3.7 (Bernstein Inequality) . *Let X_n be a random variable having binomial distribution with parameters n and p. For any $\varepsilon > 0$, we have*

$$P(|\frac{1}{n}X_n - p| > \varepsilon) \leq 2\exp(-\frac{1}{4}n\varepsilon^2).$$

Proof: We work on the $P(\frac{1}{n}X_n > p + \varepsilon)$ only.

$$\begin{aligned} P(\frac{1}{n}X_n > p+\varepsilon) &= \sum_{k=m}^{n} \binom{n}{k} p^k q^{n-k} \\ &\leq \sum_{k=m}^{n} \exp\{\lambda[k - n(p+\varepsilon)]\}\binom{n}{k} p^k q^{n-k} \\ &\leq \exp(-\lambda n\varepsilon) \sum_{k=0}^{n} \binom{n}{k} (pe^{\lambda q})^k (qe^{-\lambda p})^{n-k} \\ &= e^{-\lambda n\varepsilon}(pe^{\lambda q} + qe^{-\lambda p})^n \end{aligned}$$

with $q = 1 - p$, m the smallest integer which is larger than $n(p+\varepsilon)$ and for every positive constant λ.

It is easy to show $e^x \leq x + e^{x^2}$ for all real number x. With the help of this, we get

$$e^{-\lambda n\varepsilon}(pe^{\lambda q} + qe^{-\lambda p})^n \leq \exp(n\lambda^2 - \lambda n\varepsilon).$$

By choosing $\lambda = \frac{1}{2}\varepsilon$, we get the conclusion.

The other part can be done similarly and so we get the conclusion. $\diamondsuit$.

What we have done is, in fact, making use of the moment generating function. More skillful application of the same technique can give us even sharper bound which is applicable in more general cases. We will state, without a proof, of the sharper bound as follows:

Theorem 3.8 (Hoeffding inequality) *Let $Y_1, Y_2, \ldots, Y_n$ be independent random variables satisfying $P(a \leq Y_i \leq b) = 1$, for each i, where $a < b$. Then, for every $\varepsilon > 0$ and all n,*

$$P(\bar{Y}_n - \mathbb{E}(\bar{Y}_n) \geq \varepsilon) \leq \exp(-\frac{2n\varepsilon^2}{(b-a)^2}).$$

With this inequality, we give a very sharp bound for the size of the sample quantile.

Example 3.1 *Let $0 < p < 1$. Suppose that ξ_p is the unique solution of $F(x-) \leq p \leq F(x)$. Then, for every $\varepsilon > 0$ and all n,*

$$P(|\hat{\xi}_p - \xi_p| > \varepsilon) \leq 2\exp\{-2n\delta_\varepsilon^2\}$$

where $\delta_\varepsilon = \min\{F(\xi_p + \varepsilon) - p, p - F(\xi_p - \varepsilon)\}$.

Proof: Assignment.

The result can be stated in an even stronger way. Recall $\hat{\xi}_p$ actually depends on n. Let us now write it as $\hat{\xi}_{pn}$.

Corollary 3.1 *Under the assumptions of the above theorem, for every $\varepsilon > 0$ and all n,*

$$P(\sup_{m \geq n} |\hat{\xi}_{pm} - \xi_p| > \varepsilon) \leq \frac{2}{1-\rho_\varepsilon}\rho_\varepsilon^n.$$

where $\rho_\varepsilon = \exp(-2\delta_\varepsilon^2)$.

Remark:

1. We can choose whatever value for ε, including making it a function of n. For example, we can choose $\varepsilon = \frac{1}{\sqrt{n}}$.

2. Since the bound works for all n, we can apply it for fixed n as well as for asymptotic analysis.

We now introduce another inequality which is also attributed to Bernstein.

Theorem 3.9 (Bernstain) *Let $Y_1, \ldots, Y_n$ be independent random variables satisfying $P(|Y_i - \mathbb{E}\{Y_i\}| \leq m) = 1$, for each i, where m is finite. Then, for $t > 0$,*

$$P\Big(|\sum_{i=1}^{n} Y_i - \sum_{i=1}^{n} E\{Y_i\}| \geq nt\Big) \leq 2\exp\Big(-\frac{n^2t^2}{2\sum_{i=1}^{n} Var(Y_i) + \frac{2}{3}mnt}\Big), \tag{3.1}$$

for all positive integer n.

The strength of this inequality is at situations where m is not small but the individual variances are small.

3.5 Bahadur's representation

We have seen that the properties of the sample quantiles can be investigated through the empirical distribution function based on iid observations. This is very natural. It is very ingenious to have guessed that there is a linear relationship between the sample quantile and the sample distribution. In a not so accurate way, Bahadur showed that

$$F_n^{-1}(p) - F^{-1}(p) = C_p[F_n(\xi_p) - F(\xi_p)]$$

for some constant C_p depends on p and F when n is large. Such a result make it very easy to study the properties of sample quantiles. A key step in proving this result is to assess the size of

$$\{F_n(\xi_p + x) - F_n(\xi_p)\} - \{F(\xi_p + x) - F(\xi_p)\} \stackrel{def}{=} \Delta_n(x) - \Delta(x).$$

When x is a fixed constant, not random nor depends on n, we have

$$P\{|\Delta_n(x)-\Delta(x)|\geq t\}\leq 2\exp\{-\frac{nt^2}{2\sigma^2(x)+\frac{2}{3}t}\}$$

where

$$\sigma^2(x)=\Delta(x)\{1-\Delta(x)\}.$$

As this is true for all n, we may conclusion tentatively that

$$\Delta_n(x)-\Delta(x)=O_p(n^{-1/2}).$$

This result can be improved when x is known to be very small. Assume that the c.d.f of X satisfies the conditions

$$|\Delta(x)|=|F(\xi_p+x)-F(\xi_p)|\leq c|x|$$

for all small enough $|x|$. Let us now choose

$$x=n^{-1/2}(\log n)^{1/2}.$$

It is therefore true that $|\Delta(x)|\leq cn^{-1/2}(\log n)^{1/2}$, and $\sigma^2(x)\leq cn^{-1/2}(\log n)^{1/2}$. Applying these facts to the same inequality when n is large, we have

$$\begin{aligned}
&P\{|\Delta_n(x)-\Delta(x)|\geq 3c^{1/2}n^{-3/4}(\log n)^{3/4}\}\\
&\leq\ 2\exp\Big\{-\frac{9cn^{-1/2}(\log n)^{3/2}}{2cn^{-1/2}(\log n)^{1/2}+\frac{2}{3}c^{1/2}n^{-3/4}(\log n)^{3/4}}\Big\}\\
&\leq\ 2\exp\{-4\log(n)\}=2n^{-4}.
\end{aligned}$$

By using Borel-Cantelli Lemma, we have shown

$$\Delta_n(x)-\Delta(x)=O(n^{-3/4}(\log n)^{3/4})$$

for this choice of x, almost surely.

Now, we try to upgrade this result so that it is true uniformly for x in a small region of ξ_p.

Lemma 3.1 *Assume that the density function $f(x) \leq c$ in a neighborhood of ξ_p. Let a_n be a sequence of positive numbers such that $a_n = C_0 n^{-1/2}(\log n)^{1/2}$. We have*

$$\sup_{|x| \leq a_n} |\Delta_n(x) - \Delta(x)| = O(n^{-3/4}(\log n)^{3/4})$$

almost surely.

PROOF: Let us divide the interval $[-a_n, a_n]$ into $\alpha_n = 2n^{1/4}(\log n)^{1/2}$ equal length intervals. We round up if α_n is not an integer. Let $b_0, b_{\pm 1}, \ldots b_{\pm \alpha_n}$ be end points of these intervals with $b_0 = 0$. Obviously, the length of each interval is not longer than $C_0 n^{-3/4}$.

Let $\beta_n = \max\{F(b_{i+1}) - F(b_i)\}$ where the max is taken over the obvious range. Clearly $\beta_n = O(n^{-3/4})$.

One key observation is:

$$\sup_{|x| \leq a_n} |\Delta_n(x) - \Delta(x)| \leq \max |\Delta_n(b_i) - \Delta(b_i)| + \beta_n.$$

However, for each i, we have shown by (3.2) that

$$P\{|\Delta_n(b_i) - \Delta(b_i)| \geq 3c^{1/2} n^{-3/4}(\log n)^{3/4}\} \leq 2n^{-4}.$$

Hence, the chance for the maximum to be larger than this quantity is at most α_n times of $2n^{-4}$. In addition

$$\sum_n 2\alpha_n n^{-4}$$

which is finite. Hence, we have shown

$$\sup_{|x| \leq a_n} |\Delta_n(x) - \Delta(x)| = O(n^{-3/4}(\log n)^{3/4})$$

almost surely. This completes the proof.

To apply this result to the sample quantile, we need only show that the sample quantile will stay close to the target quantile. More precisely, we can show the following.

Lemma 3.2 *Let $0 < p < 1$. Suppose F is differentiable at ξ_p with $F'(\xi_p) = f(\xi_p) > 0$. Then, almost surely,*

$$|\hat{\xi}_p - \xi_p| \leq \frac{2(\log n)^{1/2}}{f(\xi_p) n^{1/2}}.$$

With this lemma, we are ready to claim the following.

Corollary 3.2 *Under the conditions of Lemma 3.2, we have*

$$\{F_n(\hat{\xi}_p) - F_n(\xi_p)\} - \{F(\hat{\xi}_p) - F(\xi_p)\} = O(n^{-3/4}(\log n)^{3/4})$$

almost surely.

Finally, we get the Bahadur's representation.

Theorem 3.10 *Under the conditions of Lemma 3.2, and assume that F is twice differentiable at ξ_p. Then, almost surely,*

$$\hat{\xi}_p - \xi_p = \{f(\xi_p)\}^{-1}\{p - F_n(\xi_p)\} + O(n^{-3/4}(\log n)^{3/4}).$$

PROOF: As F is twice differentiable, we have

$$F(\hat{\xi}_p) - F(\xi_p) = f(\xi_p)(\hat{\xi}_p - \xi_p) + O(\{\hat{\xi}_p - \xi_p\}^2).$$

From Lemma 3.2, we may replace F by F_n on the left hand side which will resulting an error of size $n^{-3/4}(\log n)^{3/4}$. In addition, we know that $\{\hat{\xi}_p - \xi_p\}^2 = O(n^{-1}(\log n))$. Therefore, we find

$$F_n(\hat{\xi}_p) - F_n(\xi_p) = f(\xi_p)(\hat{\xi}_p - \xi_p) + O(n^{-3/4}(\log n)^{3/4}). \tag{3.2}$$

From the definition of $\hat{\xi}_p$, we know it is either (np)th order statistic with a rounding up or down by 1 if F is continuous at all x. If so,

$$|F_n(\hat{\xi}_p) - p| \leq n^{-1}.$$

Otherwise, $\hat{\xi}_p$ is converges to ξ_p almost surely and the density $f(\xi_p)$ exists and none zero. Hence, the same bound applies almost surely. Substituting it into (4.7), we have

$$p - F_n(\xi_p) = f(\xi_p)(\hat{\xi}_p - \xi_p) + O(n^{-3/4}(\log n)^{3/4}).$$

This implies the result of this theorem.

There is a usual routine in proving asymptotic normality of a statistic. We first expand the statistics so that it consists of two terms. The first term is the sum of independent random variables. The second term is a higher order term

compared to the first one. Consequently, the classical central limit theorem is applied together with the Sluztsky's theorem.

This idea works in most cases. Badahur's representation further enhances this point of view. It finds such an expansion for a highly non-smooth function. Further, it quantifies the higher order term accurately. In statistics, we do not usually need such a precise expansion. We rarely make use of an almost sure result. However, this technique is useful.

More general results, history of the development can be found in Serfling (1980).

Problems

1. Let $F(x)$ be a one-dimensional cumulative distribution function such $F(x) = 0.5$ if and only if $x_1 < x < x_2$ for some $x_1 < x_2$. Let $\hat{\xi}_{0.5}$ be the sample median based on n iid samples from $F(x)$. Derive the limiting distribution of $\hat{\xi}_{0.5}$ when $n \to \infty$.

2. Show that under the i.i.d. and finite moment assumption, $\mathbb{E}(\bar{X}_n - \mu)^k = O(n^{-k/2})$ when k is a positive even integer; and $\mathbb{E}(\bar{X}_n - \mu)^k = O(n^{-(k+1)/2})$ when k is a positive odd integer.

3. Show that
$$s_0 = \log\left\{\frac{q(p+\varepsilon)}{p(q-\varepsilon)}\right\}$$
is the minimum point of
$$g(s) = (q + pe^s)^n \exp\{-(p+\varepsilon)s\}$$
where $p+q = 1$, $0 < p < 1$, $0 < \varepsilon < q$.
Show that
$$\left(\frac{q}{q-\varepsilon}\right)^{(q-\varepsilon)} \left(\frac{p}{p+\varepsilon}\right)^{(p+\varepsilon)} \leq \exp(-2\varepsilon^2).$$

4. Let $X_1, X_2, \ldots, X_n$ be i.i.d. random variables. Calculate
$$\mathbb{E}(\bar{X}_n - \mu)^k$$
for $k = 3, 4, 5$ as functions of the moments of X_1.

5. Let $F_n(t)$ be the empirical distribution function based on uniform[0, 1] i.i.d. random variables. Show that

$$\sup_{0\le t\le 1}|F_n(t)-t| = \sup_{0\le p\le 1}|F_n^{-1}(p)-p|.$$

6. Let $\hat{\xi}_{0.25}$ and $\hat{\xi}_{0.75}$ be empirical 25% and 75% quantiles. Derive the limiting distribution of

$$\sqrt{n}[(\hat{\xi}_{0.75}-\hat{\xi}_{0.25})-(\xi_{0.75}-\xi_{0.25})]$$

under conditions comparable to the conditions in Theorem 3.10 on the distribution function F. Discuss how can we have the derivative conditions on F weakened?

7. Prove Example 3.1.

8. Prove Lemma 3.2

9. The residual in the Bahadur's representation has higher order if it is in the sense of "in probability", not "almost surely". Show that the order can be improved to $O_p(n^{-3/4}(\log n)^{1/2})$.

Chapter 4

Smoothing method

In some statistical applications, the parameter to be estimated is a function on some Euclidean space, not a real number or real vector. The number of observations available to estimate the value of this function at any given point is often conceptually 0. A simple example is to estimate the density function of an absolutely continuous random variable. It is widely believed that, with a reference yet to be found, there does not exist an unbiased estimator in general for density function. A related problem is on non-parametric regression analysis. In this case, the objective is to make inference on the regression function. In theory, a response variable y can often be predicted in terms of a few covariates or design variables x. The regression function is the conditional expectation of Y given $X = x$. When X is random and has continuous distribution, the number of observations of Y at exactly $X = x$ is 0. One again has to make use of observed responses corresponding to X values in a neighborhood of x. We will work on density estimation first.

4.1 Kernel density estimate

Let $F(x)$ be the common cumulative distribution function of an iid sample on real numbers. When the density function is smooth, we have

$$f(x) \approx \frac{F(x+h) - F(x-h)}{2h}$$

when h is a small number. Since $F_n(x)$ is a good estimator of $F(x)$, we may hence estimate $f(x)$ by

$$\hat{f}(x) = \frac{F_n(x+h) - F_n(x-h)}{2h} = \frac{1}{2nh} \sum_{i=1}^{n} \mathbb{1}(|X_i - x| \leq h)$$

with a proper choice of small h. It is seen that this estimator is the ratio of the average number of observations falling into the interval $[x-h, x+h]$ to the length of the interval. When h is large, the average may not reflect the density at x but the average density over a short interval containing x. Thus, the estimator may have large bias. When h is very small, the number of observations in the interval will fluctuate from sample to sample more severely, thus the estimator has large variance. In general, we choose the bandwidth h according to the sample size. As it will be seen, the basic requirements include $h \to 0$ and $nh \to \infty$ as the sample size $n \to \infty$. Because of this, most quantities to be discussed are functions of n even if there is no explicit notational indication.

Let

$$K(x) = \frac{1}{2} I(-1 < x < 1)$$

which is itself a density function and $K_h(x) = h^{-1}K(x/h)$. We can then write

$$\hat{f}(x) = n^{-1} \sum_{i=1}^{n} K_h(X_i - x) = \int K_h(t-x) dF_n(t). \tag{4.1}$$

Hence, the density estimator is the average value of $K_h(X_i - x)$. The estimator defined by (4.1) is called the kernel density estimator. We call K the kernel of this density estimator and h the bandwidth.

In this type of estimators, the contribution of X_i toward the density estimate at x is determined by $K_h(X_i - x)$. It is easy to see that we may replace $K(x)$ by any other density function, and the resulting $\hat{f}(x)$ is still a sensible density estimator. With the previous choice of K, while observations within $\pm h$ neighborhood of x have equal contribution, the observations out side of this interval contribute nothing to the density. It is more reasonable to make $K(x)$ a smoothly decreasing function in $|x|$. Thus, a popular choice of $K(x)$ is the density function of the standard normal distribution. Although the rest of our discussion can be easily generalized to multi-dimensional densities, the presentation is the simplest when

X is a scaler. Unless otherwise specified, we assume X is a scale in the rest of the chapter.

Some basic conditions on the kernel function and the density functions are as follows.

1. $K(x)$ is a density function.
2. $\int sK(s)ds = 0$.
3. $\mu_2 = \int s^2 K(s)ds < \infty$.
4. $R(K) = \int K^2(s)ds < \infty$.
5. $\lim_{s\to\pm\infty} K(s) = 0$.
6. $f(x)$ is continuously differentiable to the second order and has bounded second derivative.

The above conditions can be relaxed to obtain the results we are set to discuss. It is not hard to verify that the density function of the standard normal distribution satisfies all the conditions listed. Hence, the normal density function can be used as a kernel function, which results in a kernel density estimator with the properties to be presented.

4.1.1 Bias of the kernel density estimator

A simple expression of the bias of the kernel density estimator can be easily obtained. Note that due to iid structure of the data,

$$E[\hat{f}(x)] = \int K_h(t-x)f(t)dt = \int K(s)f(x+hs)ds.$$

When $f(x)$ is continuous at x, we have

$$E[\hat{f}(x)] \to f(x)$$

when $h \to 0$ and $K(x)$ is a density function.

Assume the conditions on the density function $f(x)$ and $K(x)$ are all satisfied. Then

$$f(x+hs) = f(x) + f'(x)hs + \frac{h^2}{2} f''(\xi)s^2$$

for some ξ between x and $x+hs$. Under these conditions, we have

$$E[\hat{f}(x)] = f(x) + \frac{h^2}{2} \int s^2 K(s) f''(\xi) ds.$$

Since $f''(x)$ is continuous and bounded, we have

$$\int s^2 K(s) f''(\xi) ds \to \mu_2 f''(x)$$

as $h \to 0$. Consequently, we have

$$E[\hat{f}(x)] = f(x) + \frac{\mu_2 h^2}{2} f''(x) + o(h^2).$$

In conclusion, the bias of a kernel density estimator is typically in the order of h^2 when, for example, the kernel function is chosen to be symmetric, and the density function has continuous, bounded second derivative.

We now look into the variance properties of the kernel density estimation.

4.1.2 Variance of the kernel density estimator

Due to the iid structure in the kernel density estimator, we have

$$\text{VAR}(\hat{f}(x)) = n^{-1} \text{VAR}(K_h(X-x))$$

where X is a generic random variable whose density function is given by $f(x)$. It is easily verifiable that

$$\text{VAR}(K_h(X-x)) \leq \mathbb{E}[K_h(X-x)]^2.$$

The interesting part is that the difference is $\{\mathbb{E}[K_h(X-x)]\}^2$ which equal $\{f(x) + O(h^2)\}^2$ according to the result in the last section. At the same time, it will be seen that $\mathbb{E}[K_h^2(X-x)] = O(h^{-1})$ which tends to infinity as $h \to 0$. Hence, the leading term of the variance is determined purely by $\mathbb{E}[K_h^2(X-x)]$.

Similar to the bias computation, we have

$$\begin{aligned}\mathbb{E}[K_h^2(X-x)] &= \int K_h^2(t-x)f(t)dt \\ &= h^{-1}\int K^2(s)f(x+hs)ds \\ &= h^{-1}f(x)R(K)ds(1+o(1)).\end{aligned}$$

Hence, we have

$$\text{VAR}(\hat{f}(x)) = (nh)^{-1}f(x)R(K)(1+o(1)). \tag{4.2}$$

It is seen then that the mean squared error of the kernel density estimator is

$$\text{MSE}(\hat{f}(x)) = (nh)^{-1}f(x)\|K\|^2 + h^4\mu_2^2[f''(x)]^2/4 + o((nh)^{-1}+h^4).$$

Thus, in order for $\hat{f}(x)$ to be consistent, a set of necessary conditions on h are

$$h \to 0, \;\; nh \to \infty.$$

To minimize the order of MSE as $n \to \infty$, we should choose $h = n^{-1/5}$.

The best choices of h at different x are not the same. Thus, one may instead search for h such that the integrated MSE is minimized. For this purpose, we further assume $\int [f''(x)]^2 dx < \infty$ and the integration of the remainder term remains to the order before the integration. If so, we have the mean integrated squared error as

$$\text{MISE} = (nh)^{-1}\|K\|^2 + h^4\mu_2^2\int [f''(x)]^2 dx/4 + o((nh)^{-1}+h^4). \tag{4.3}$$

The optimal choice of h is then

$$h_{opt} = \left[\frac{\|K\|^2}{n\mu_2^2\int [f''(x)]^2 dx}\right]^{1/5}.$$

With this h, we have

$$\text{MISE}_{opt} = \frac{5}{4}\{\mu_2^2\|K\|^4\int [f''(x)]^2 dx\}^{1/5} n^{-4/5}.$$

In this expression, we cannot make $\int [f''(x)]^2 dx$ change as this comes with the data. We have some freedom to find a K to minimize the MISE. This is equivalent to minimize

$$\mu_2^2 \|K\|^4. \tag{4.4}$$

This quantity does not depend on the choice of the bandwidth h. That is, if the kernel function K minimizes (4.4), K_h also minimizes it as long as $h > 0$.

The solution to this minimization problem is found to be

$$K(x) = \frac{3}{4}(1-x^2)^+$$

which is called Epanechnikov kernel. It is found, however, other choices of K do not loss much of the efficiency. For example, choosing normal density function as the kernel function is 95% efficient. It means that one need about 5% more sample to achieve the same level of precision if the normal density function is used as kernel rather than the optimal Epanechnikov kernel is applied.

In conclusion, the choice of K is not considered very important. The choice of the bandwidth parameter h is. There are thousands of papers published on this topic. We do not intend to carry you away here.

4.1.3 Asymptotic normality of the kernel density estimator

In most statistical research papers, we are interested in finding constant sequences a_n, b_n such that

$$a_n[\hat{f}(x) - f(x) - b_n] \xrightarrow{d} Y$$

for some non-degenerate random variable Y. Such a result can then be used to construct confidence bounds for $f(x)$ or perform hypothesis test. Since $\hat{f}(x)$ has an iid structure, this Y must have normal distribution.

Denote $Z_{ni} = K_h(X_i - x) - \mathbb{E}[K_h(X_i - x)]$, then $\hat{f}(x) - \mathbb{E}[\hat{f}(x)] = n^{-1}\sum_{i=1}^n Z_{ni}$ which is asymptotically normal when the Liapunov condition is satisfied. It is easy to verify that the Liapunov condition is satisfied when for some $\delta > 0$,

$$\int K^{2+\delta}(s)ds < \infty; \text{ and } f(x) > 0. \tag{4.5}$$

Let $\sigma_n^2 = Var(Z_{n1})$. Under (4.5) and other conditions specified earlier, we have

$$\frac{1}{\sqrt{n}\sigma_n}\sum_{i=1}^{n} Z_{ni} \xrightarrow{d} N(0,1).$$

In general, we prefer to know the limiting distribution of $\hat{f}(x) - f(x)$ rather than that of $\hat{f}(x) - \mathbb{E}[\hat{f}(x)]$ after properly scaled. The above result helps if

$$\frac{1}{\sqrt{n}\sigma_n} n[\mathbb{E}\{K_h(X-x)\} - f(x)]$$

converges to some constant. Recall that $\sigma_n^2 = f(x)R^2(K)h^{-1} + o(h^{-1})$ and $\mathbb{E}[\{K_h(X-x)\} - f(x)] = \mu_2 f''(x)h^2/2 + o(h^2)$. Hence,

$$\frac{1}{\sqrt{n}\sigma_n} n[\mathbb{E}\{K_h(X-x)\} - f(x)] = O(n^{1/2}h^{1/2}h^2) = O(n^{1/2}h^{5/2}).$$

By choosing $h = n^{-1/5}$, the limiting distribution result becomes

$$\frac{1}{\sqrt{nh}}[\hat{f}(x) - f(x)] \to N(a, \sigma^2)$$

with $a = \mu_2[f''(x)]/2$ and $\sigma^2 = f(x)R(K)$.

4.2 Non-parametric regression analysis

A related problem in statistics is the non-parametric regression analysis. The data in such applications consist of pairs (X_i, Y_i), $i = 1, 2, \ldots, n$ from some probability model. It is desirable to use X as predictor to predict the response value Y. From probability theory, $\mathbb{E}(Y|X)$ minimizes

$$\mathbb{E}[Y - g(X)]^2$$

among all measurable function of X. In statistical literature, $m(X) = \mathbb{E}(Y|X)$ is also called the regression function of Y with respect to X.

In other applications, x values in the model are selected by a design. Thus, they are not random. A commonly assumed model for the data in this situation is

$$Y_i = m(x_i) + v^{1/2}(x_i)\varepsilon_i \tag{4.6}$$

for $i = 1,2,\ldots,n$, where x_i's are design points and $m(x)$ is the unknown non-parametric regression function, $v(x)$ is the variance function, and ε_i are random error. If $v(x)$ is a constant function, the model is homoscedastic, otherwise, it is heteroscedastic. In both cases, random design or fixed design, our observations consist of n pairs of (Y_i, X_i), $i = 1,2,\ldots,n$.

4.2.1 Kernel regression estimator

Intuitively, $m(x) = \mathbb{E}(Y|X = x)$ is the average value of Y given $X = x$. When X is random and has continuous distribution, the event $X = x$ has probability zero. Thus, in theory, the number of observations of (X_i, Y_i) in the sample such that $X_i = x$ for any x is practically zero. It is impossible to estimate $m(x)$ for any given x unbiasedly. When it is reasonable to believe that $m(x)$ is a continuous function of x, however, one may collect information in a small neighborhood of x for the purpose of estimating $m(x)$. Consider such a small neighborhood $[x-h, x+h]$, the average value of Y_i correspond to x_i's in this interval is

$$\sum_{i=1}^{n} y_i \mathbb{1}(|X_i - x| \leq h) / \sum_{i=1}^{n} \mathbb{1}(|X_i - x| \leq h) = \sum_{i=1}^{n} y_i K_h(X_i - x) / \sum_{i=1}^{n} K_h(X_i - x)$$

where $K(x) = \frac{1}{2}\mathbb{1}(|x| \leq 1)$. Clearly, $K(x)$ can be replaced by any other density function of x to get a general kernel regression estimator:

$$\hat{m}(x) = \sum_{i=1}^{n} y_i K_h(X_i - x) / \sum_{i=1}^{n} K_h(X_i - x). \tag{4.7}$$

Both K and h play the same roles as in the kernel density estimate.

When X is random and has absolutely continuous distribution, we may estimate the joint density function of (X, Y) by a kernel density estimator

$$\hat{f}(x,y) = n^{-1} \sum_{i=1}^{n} K_h(y_i - y) K_h(x_i - x).$$

In practice, we may choose two different kernel functions K and two unequal bandwidths h. We could also replace $K(y)K(x)$ by a density function of a random vector with two dimension. The theory we intend to illustrate will not change. Thus, we will only discuss the problem under the above simplistic setting.

The conditional density function of Y given $X = x$ is naturally estimated by

$$\hat{f}(y|x) = \hat{f}(x,y)/\hat{f}(x) \tag{4.8}$$

where $\hat{f}(x)$ is the kernel density estimator of $f(x)$ with kernel function K and the same bandwidth h. It is seen that

$$\int y\hat{f}(y|x)dy = \hat{m}(x) \tag{4.9}$$

assuming the range of y is $(-\infty,\infty)$. Otherwise, one can choose the kernel function K with compact support to ensure the validity of the equality. The above identity is then true for y not on the boundary, and when h is very small.

We have seen that the kernel regression estimator $\hat{m}(x)$ is well motivated both for random and non-random X.

4.2.2 Local polynomial regression estimator

The kernel density estimator can be generalized or be regarded as a special case of another method. In any small neighborhood of a point x_0, one may fit a polynomial model to the data:

$$m(x) = \beta_0 + \beta_1(x-x_0) + \cdots + \beta_p(x-x_0)^p$$

for some integer $p \geq 0$. If $m(x)$ is a smooth function, this model is justified by Taylor's expansion at least for x-values in a small neighborhood of x_0. Let $N(x_0;h)$ be small neighborhood of x_0 indexed by h. One can then estimate $m(x)$ by the least sum of squares based on data in $N(x_0,h)$. For a given p, we search for $\beta_0,\ldots,\beta_p$ to minimize

$$\sum_{x_i \in N(x_0;h)} [Y_i - \{\beta_0 + \beta_1(x-x_0) + \cdots + \beta_p(x-x_0)^p\}]^2.$$

Instead of defining a neighborhood directly, one may use a kernel function to reduce the weights of observations at x_i which are far away from x_0. Employing the same idea as in the kernel regression estimator, we select a suitable kernel function K and replace the above sum of squares by the sum of weighted squares:

$$\sum_{i=1}^{n} K_h(x_i-x_0)[Y_i - \{\beta_0 + \beta_1(x-x_0) + \cdots + \beta_p(x-x_0)^p\}]^2.$$

We then estimate $m(x_0)$ by $\hat{\beta}_0$. When we choose $p = 0$, this estimator reduces to the kernel regression estimator. Recently statistical literature indicates that the local polynomial regression method has some superior properties. It is, however, too much material for us to cover much of them in this course. Thus, we will only study the case when $p = 0$.

4.2.3 Asymptotic bias and variance for fixed design

Let us first consider the case when x_i's are not random and are equally spaced in the unit interval [0, 1]. That is, let us assume that $x_i = n^{-1}(i - 1/2)$ for $i = 1, 2, \ldots, n$. Under model (4.6) and by the definition of (4.7), we have

$$\mathbb{E}[\hat{m}(x)] = \sum_{i=1}^{n} m(x_i) K_h(x_i - x) / \sum_{i=1}^{n} K_h(x_i - x).$$

At the same time, by the mean value theorem for integrals, we have

$$\begin{aligned} \int_0^1 m(t) K_h(t-x) dt &= \sum_{i=1}^{n} \int_{(i-1)/n}^{i/n} m(t) K_h(t-x) dt \\ &= \sum_{i=1}^{n} m(t_i) K_h(t_i - x) \end{aligned}$$

for some t_i's in $[(i-1)/n, i/n]$. Thus, we have

$$\begin{aligned} & \Big| \int_0^1 m(t) K_h(t-x) dt - n^{-1} \sum_{i=1}^{n} m(x_i) K_h(x_i - x) \Big| \\ \leq\ & (nh)^{-1} \sum_{i=1}^{n} |m(x_i) K_h(x_i - x) - m(t_i) K_h(t_i - x)| \\ =\ & O((nh^2)^{-1}) \end{aligned}$$

when $m(x)$ and $K(x)$ both have bounded first derivatives. If K has compact support, and x is an interior point, the range of the summation or integration can be restricted to an interval of length proportional to h. Consequently, the order assessment can be refined to $O((nh)^{-1})$. At the same time, when x is an inner point of the unit interval,

$$\int_0^1 m(t) K_h(t-x) dt = \int K(s) m(x + hs) ds = m(x) + \frac{\mu_2 m''(x)}{2} h^2 + o(h^2).$$

Hence,

$$n^{-1}\sum_{i=1}^{n} m(x_i)K_h(x_i-x) = m(x) + \frac{\mu_2 m''(x)}{2}h^2 + o(h^2) + O((nh^2)^{-1}).$$

Using the same technique, we have

$$\sum_{i=1}^{n} K_h(x_i-x) = 1 + o(h^2) + O((nh^2)^{-1}).$$

Combined, we have

$$\mathbb{E}[\hat{m}(x)] = m(x) + \frac{\mu_2 m''(x)}{2}h^2 + o(h^2) + O((nh^2)^{-1}).$$

The order assessment here is slightly different from the literature, for example, page 122 of Wand and Jones (1995). One may investigate more closely on the order of the error when we approximate the summation with integration to find if ours is not precise enough.

The computation of the asymptotic variance is done in the similar fashion. We have

$$\begin{aligned} \text{VAR}(\hat{m}(x)) &= \sum_{i=1}^{n} v(x_i)K_h^2(x_i-x)/[\sum_{i=1}^{n} K_h(x_i-x)]^2 \\ &= (nh)^{-1}v(x)R(K) + o(h^2+(nh)^{-1}). \end{aligned}$$

Again, the order assessment here is different from some standard literature.

4.2.4 Bias and variance under random design

When X is random, the bias of the kernel regression estimator is harder to determine if we interpret the bias very rigidly. The main problem is from the fact that the kernel regression estimator is a ratio of two random variables. It is well known that for any two random variables X and Y, it is usually true that $\mathbb{E}[X/Y] \neq \mathbb{E}(X)/\mathbb{E}(Y)$. When Y takes a value near 0, the ratio becomes very unstable. The unstable problem does not get much better even if the chance for Y close to 0 is very small.

To avoid this problem, we adopt a notion of the asymptotic bias and variance. Suppose $a_n^{-1}(Z_n - b_n) \to Z$ in distribution such that $\mathbb{E}(Z) = 0$ and $Var(Z) = 1$. We

define the asymptotic mean and variance of Z_n as b_n and a_n^2. A similar definition has been given in Shao (1998), however, this definition has not appeared anywhere else to my best knowledge.

The numerator $U_n = \sum_{i=1}^n y_i K_h(X_i - x)$ and the denominator $V_n = \sum_{i=1}^n K_h(X_i - x)$ in $\hat{m}(x)$ are both sum of iid random variables. We look for proper constants a_n and (u_n, v_n) such that

$$a_n[(U_n - u_n, V_n - v_n)]$$

has limiting distribution. For this purpose, we first search for approximate means and variances of U_n and V_n.

It is seen that

$$\begin{aligned}
\mathbb{E}[U_n] &= \mathbb{E}[\sum_{i=1}^n m(X_i)K(X_i - x)] \\
&= n\int m(t)K_h(t-x)f(t)dt \\
&= n\int m(x+sh)f(x+sh)K(s)ds \\
&= n[m(x)f(x) + \frac{1}{2}\{m''(x)f(x) + 2m'(x)f'(x) + m(x)f''(x)\}\mu_2 h^2] \\
&\quad + o(nh^2). \qquad (4.10)
\end{aligned}$$

Thus, we put

$$u_n = n[m(x)f(x) + \frac{1}{2}\{m''(x)f(x) + 2m'(x)f'(x) + m(x)f''(x)\}\mu_2 h^2].$$

Similarly, we have

$$\text{VAR}[U_n] = \mathbb{E}\big[\sum_{i=1}^n v(X_i)K_h^2(X_i - x)\big] + \text{VAR}\big[\sum_{i=1}^n m(X_i)K_h(X_i - x)\big]. \qquad (4.11)$$

It is seen that

$$\begin{aligned}
\mathbb{E}[v(X)K_h^2(X-x)] &= \int v(t)K_h^2(t-x)f(t)dt \\
&= h^{-1}v(x)f(x)R(K) + o(h^{-1}).
\end{aligned}$$

For the second term in (4.10), we have

$$\begin{aligned}
&\mathrm{VAR}[\sum_{i=1}^{n} m(X_i)K_h(X_i-x)] \\
&= n\mathrm{VAR}[m(X)K_h(X_i-x)] \\
&= n[\mathbb{E}\{m^2(X)K_h^2(X-x)\} - \{\mathbb{E}m(X)K_h(X-x)\}^2] \\
&= n\mathbb{E}\{m^2(X)K_h^2(X-x)\} + O(n).
\end{aligned}$$

Further,

$$\begin{aligned}
\mathbb{E}\{m^2(X)K_h^2(X-x)\} &= \int m^2(t)K_h^2(t-x)f(t)dt \\
&= h^{-1}\int m^2(x+sh)f(x+sh)K(s)ds \\
&= h^{-1}m^2(x)f(x)R(K) + o(h^{-1}).
\end{aligned}$$

In conclusion, we have shown

$$\mathrm{VAR}(U_n) = nh^{-1}\{v(x)+m^2(x)\}f(x)R(K) + o(nh^{-1}).$$

Using similar calculation, we have

$$\begin{aligned}
\mathbb{E}[V_n] &= n[f(x) + \frac{1}{2}f''(x)\mu_2 h^2] + o(nh^2), \\
\mathrm{VAR}(V_n) &= nh^{-1}f(x)R(K) + o(nh^{-1}).
\end{aligned}$$

In view of the above bias and variance results, it is apparent that we should choose $h^{-1/5}$ and therefore $a_n = n^{-(3/5)}$ to produce some meaningful limiting distribution. Assume that the conditions for the joint asymptotic normality of $n^{-3/5}[U_n - \mathbb{E}(U_n), V_n - \mathbb{E}(V_n)]$ are satisfied. For convenience, write $\bar{U}_n = n^{-1}U_n$ and so on for the sake that $\bar{U}_n \to m(x)f(x)$ in probability. It makes the following presentation simpler. We have

$$n^{2/5}[\bar{U}_n - \bar{u}_n, \bar{V}_n - \bar{v}_n] \to N(0,\Delta) \tag{4.12}$$

for covariance matrix Δ consists of $\delta_{11} = \{v(x)+m^2(x)\}f(x)R(K)$, $\delta_{22} = f(x)R(K)$ and $\delta_{12} = m(x)f(x)R(K)$. The computation of δ_{12} is left out as an assignment problem.

Finally, we have

$$\bar{U}_n/\bar{V}_n - m(x) = [\bar{U}_n/\bar{V}_n - \bar{u}_n/\bar{v}_n] + [\bar{u}_n/\bar{v}_n - m(x)].$$

and

$$[\bar{u}_n/\bar{v}_n - m(x)] = \frac{1}{2}\{m''(x) + 2\mu_2 m'(x)f'(x)/f(x)\}h^2 + o(h^2).$$

With that, we have

$$\begin{aligned} n^{2/5}\{\hat{m}(x) - m(x)\} &= n^{2/5}\left[\frac{\bar{U}_n}{\bar{V}_n} - m(x)\right] \\ &= \{m''(x) + 2m'(x)f''(x)/f(x)\mu_2\} \\ &\quad + \frac{n^{2/5}[\{\bar{U}_n - \bar{u}_n\}\bar{v}_n + \bar{u}_n\{\bar{v}_n - \bar{V}_n)\}]}{V_n v_n} + o_p(1) \\ &= \{m''(x) + 2m'(x)f''(x)/f(x)\mu_2\} \\ &\quad + \frac{n^{2/5}[\{\bar{U}_n - \bar{u}_n\}\bar{v}_n + \bar{u}_n\{\bar{v}_n - \bar{V}_n)\}]}{v_n^2} + o_p(1). \end{aligned}$$

It is then obvious that

$$n^{2/5}\{\hat{m}(x) - m(x)\} \to N(a, \sigma^2),$$

with $a = \frac{\mu_2}{2}\{m''(x) + 2m'(x)f'(x)/f(x)\}$ and

$$\begin{aligned} \sigma^2 &= \frac{f^2(x)\delta_{11} + 2m(x)f^2(x)\delta_{12} + m^2(x)f^2(x)\delta_{22}}{f^4(x)} \\ &= v(x)R(K)\{f(x)\}^{-1}. \end{aligned} \tag{4.13}$$

Because of this, it is widely cited that the asymptotic bias of $\hat{m}(x)$ is given by

$$\frac{1}{2}\{m''(x) + 2m'(x)f''(x)/f(x)\mu_2\}h^2$$

and the asymptotic variance is given by

$$(nh)^{-1}v(x)R(K)\{f(x)\}^{-1}.$$

The citation is not entirely true as two important conditions cannot be ignored: (1) $K(x)$ has compact support; (2) the bandwidth parameter $h = O(n^{-1/5})$.

4.3 Assignment problems

1. Verify that the Liapunov condition is satisfied when (4.5) is met in addition to other conditions on K and f specified before (4.5).

2. Show that when the second moments of X and Y exist,

$$\mathbb{E}[Y - g(X)]^2$$

   is minimized among all possible choice of measurable function of X when $g(X) = \mathbb{E}\{Y|X\}$.

3. Verify the order assessment given in (4.10). Present your own result if your assessment is different.

4. Prove that (4.9) is true as defined in the content. Why is it necessary to assume that the range of x is the entire space of real numbers? List two meaningful generalizations of this result (which is too restrictive in many ways).

5. Verify the result on the variance covariance matrix Δ defined in (4.12).

6. Verify the result given in (4.13)

Chapter 5

Asymptotic Results in Finite Mixture Models

5.1 Finite mixture model

In statistics, a model means a family of probability distributions. Given a random variable X, its cumulative distribution function (c.d.f.) is defined to be $F(x) = P(X \leq x)$ for $x \in \mathbb{R}$. Let $p \in (0,1)$ and

$$F(x) = \sum_{0 \leq k \leq x} \binom{n}{k} p^k (1-p)^{n-k}$$

for $x \in \mathbb{R}$ and the summation on k is over integers. A random variable with its c.d.f. having the above algebraic expression is known as binomially distributed, or it is a binomial random variable. It contains two parameters n and p. A particular pair of values in n and p gives one particular binomial distribution. The binomial distribution family is the collection of binomial distributions with all possible parameter values in n and p. We may form a narrower binomial distribution family by holding n fixed. Whether or not n is fixed, this distribution family is regarded as *Binomial Model*.

A discrete integer valued random variable has its p.m.f. given by

$$f(x) = P(X = x) = F(x) - F(x-1)$$

for $x = 0, \pm 1, \ldots$. For a binomial X, its p.m.f. is given by

$$\text{BIN}(k; n, p) = P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

for $k = 0, 1, \ldots, n$. We write $X \sim \text{BIN}(n, p)$. Let π be a value between 0 and 1 and let

$$f(k) = \pi \text{BIN}(k; n, p_1) + (1-\pi)\text{BIN}(k; n, p_2) \tag{5.1}$$

for $k = 0, 1, \ldots, n$ with some parameters n and $p_1, p_2 \in [0, 1]$. Clearly, the above $f(\cdot)$ is also a p.m.f. The distributions whose p.m.f. have algebraic structure (5.1) form a new distribution family. Because $f(\cdot)$ is a convex combination of two p.m.f. 's from a well known distribution family, its distribution is called a binomial mixture distribution. We subsequently have a binomial mixture model.

Be aware that we use $f(\cdot)$ very loosely as a general symbol for a p.m.f. or a p.d.f. Its specifics may change from one paragraph to another. We must not interpret it as a specific p.m.f. or p.d.f. with great care.

Let $\{f(x; \theta) : \theta \in \Theta\}$ be a parametric model and $G(\theta)$ be a c.d.f. on Θ. We obtain a mixture distribution represented by

$$f(x; G) = \int_\Theta f(x; \theta) dG(\theta), \tag{5.2}$$

where the integration is understood in the Lebesgue-Stieltjes sense. When $G(\cdot)$ is absolutely continuous with density function $g(\theta)$, the integration equals $\int_\Theta f(x; \theta) g(\theta) d\theta$. We are often interested in the situation where G is a discrete distribution assigning positive probabilities $\pi_1, \ldots, \pi_m$ to finite number of θ-values, $\theta_1, \ldots, \theta_m$ such that $\sum_{j=1}^m \pi_j = 1$. In this case, the mixture density (5.2) is reduced to the familiar convex combination:

$$f(x; G) = \sum_{j=1}^m \pi_j f(x; \theta_j). \tag{5.3}$$

Because a distribution is also referred as a population in some context, we therefore also call $f(x; \theta_j)$ a sub-population of the mixture population. We call θ_j sub-population parameter and π_j the corresponding mixing proportion. When all $\pi_j > 0$, these component parameters $\{\theta_1, \ldots, \theta_m\}$ are the support points of G. The order of the mixture model is m if G has at most m support points.

The density functions $f(x; G)$ in (5.2) form a mixture distribution family, and therefore a mixture model. The density functions $f(x; G)$ in (5.3) form a finite

mixture distribution family, and therefore a finite mixture model. The collection of the mixing distribution will be denoted as $\mathbb{G}$. A mixture model is a distribution family characterized by

$$\{f(x;G) = \int_{\Theta} f(x;\theta)dG(\theta) : G \in \mathbb{G}\}$$

which requires both $\{f(x;\theta) : \theta \in \Theta\}$ and $\mathbb{G}$ fully specified.

We will use $F(x;\theta)$ as the c.d.f. of the component density function $f(x;\theta)$ and similarly for $F(x;G)$ and $f(x;G)$. The same symbols F and f are used for both component and mixture distributions, we have to pay attention to the symbol in their entry. We also use $G(\{\theta\})$ for the probability the distribution G assigns to a specific θ value and similarly for $F(\{x\})$. We may also refer $f(x;G)$ as a mixture density, a mixture distribution or a mixture model.

We have now completed the introduction of the mixture model.

5.2 Test of homogeneity

Finite mixture models are often used to help determine whether or not data were generated from a homogeneous or heterogeneous population. Let $X_1,\ldots,X_n$ be a sample from the mixture p.d.f.

$$(1-\gamma)f(x,\theta_1)+\gamma f(x,\theta_2), \tag{5.4}$$

where $\theta_1 \leq \theta_2 \in \Theta$ and $0 \leq \gamma \leq 1$. We wish to test the hypothesis

$$H_0 : \theta_1 = \theta_2, \text{ (or equivalently } \gamma = 0, \text{ or } \gamma = 1).$$

Namely, we test whether or not the observations come from a homogeneous population $f(x,\theta)$.

In applications, the null model is the default position. That is, unless there is a strong evidence in contradiction, the null model is regarded as "true". At the same time, searching for evince against the null model in favour of a specific type of alternative is a way to establishing the new theory. We do not blindly trust a new theory unless it survives serious challenges.

There are many approaches to this hypothesis test problem. Given the nice i.i.d. structure and the parametric form of the finite mixture model, the classical likelihood ratio test will be the one to be discussed.

5.3 Binomial mixture example

Consider the situation where the kernel distribution is Binomial with known size parameter m and probability of success parameter θ. Let $X_1, \ldots, X_n$ be a set of i.i.d. random variables with common finite mixture of binomial distributions such that

$$P(X = k) = (1-\pi)\text{BIN}(k; m, \theta_1) + \pi\text{BIN}(k; m, \theta_2)$$

where π and $1-\pi$ are mixing proportions, and θ_1 and θ_2 are component distribution parameters. Clearly, the parameter space of the mixing parameters θ_1 and θ_2 are bounded. The likelihood ratio statistic is stochastically bounded. We now first demonstrate this fact.

Let n_k be the number of observed $X_i = k$ for $k = 0, 1, \ldots, m$. The log-likelihood function is given by

$$\ell_n(\pi, \theta_1, \theta_2) = \sum_{k=0}^{m} n_k \log\{(1-\pi)f(k; m, \theta_1) + \pi f(k; m, \theta_2)\}$$

Let $\hat{\theta}_k = n_k/n$. By Jensen's inequality, we have

$$\ell_n(\pi, \theta_1, \theta_2) \leq n \sum_{k=0}^{m} \hat{\theta}_k \log \hat{\theta}_k$$

for any choices of π, θ_1 and θ_2. Let $p_k^* = P(X = k)$ under the true distribution of X. It is well known that in distribution, we have

$$n \sum_{k=0}^{m} \hat{p}_k \log \hat{p}_k - n \sum_{k=0}^{m} \hat{p}_k \log p_k^* \to \chi_m^2$$

as $n \to \infty$. Let us also use M_n for the likelihood ratio statistic. We find

$$M_n = 2\{\sup \ell_n(\pi, p_1, p_2) - \sup \ell_n(1, p, p)\} \leq n \sum_{k=0}^{m} \hat{p}_k \log \hat{p}_k - n \sum_{k=0}^{m} \hat{p}_k \log p_k^*$$

which has a χ_m^2 limiting distribution. That is, $M_n = O_p(1)$, or it is stochastically bounded.

One should have realized that the conclusion $M_n = O_p(1)$ does not particularly rely on X_i's having a binomial distribution. The conclusion remains true when X_i's have common and finite number of support points.

In spite of boundedness of M_n for binomial mixtures, finding the limiting distribution of M_n troubled statisticians as well as geneticists for a long time. The first most satisfactory answer is given by Chernoff and Lander(1985JSPI). Unlike the results under regular models, the limiting distribution of the likelihood ratio statistic under binomial mixtures is not an asymptotic pivotal. The outcomes vary according to the size of m, and true null distribution p and so on. We now use the simplest case with $m = 2$ special component parameter values for illustration. In this case,

$$H_0 : \ f(k;2,0.5), \ \ H_a : \ (1-\pi)f(k;2,0.5) + \pi f(k;2,0.5+\theta).$$

We investigate the limiting distribution of the likelihood ratio statistic in the presence of n i.i.d. observations from H_0.

Under the alternative hypothesis H_a, the parameter space of (π, θ) is a unit square $[0,1] \times [0,1]$. The null hypothesis is made of two lines in this unit square: one is formed by $\pi = 0$ and the other is $\theta = 0.5$. Unlike the hypothesis test problems under regular models, all points on these two lines parameterize the same distribution. The derivation of new limiting distribution naturally starts from how to avoid this non-identiability. Chernoff and Lander were the first to use parameter transformation.

Using the parameter setting under the alternative model, we note

$$\begin{aligned} P(X=0) &= 0.25(1-\pi) + \pi(1-\theta)^2; \\ P(X=1) &= 0.5(1-\pi) + 2\pi\theta(1-\theta); \\ P(X=2) &= 0.25(1-\pi) + \pi\theta^2. \end{aligned}$$

If the data are from the null model, parameters π, θ are not uniquely defined and therefore cannot be consistently estimated. At the same time, let

$$\xi_1 = 0.5\pi(\theta - 0.5); \ \ \xi_2 = \pi(\theta - 0.5)^2.$$

The null model is uniquely defined by $\xi_1 = \xi_2 = 0$. After the parameter transformation, we find

$$\begin{aligned} P(X=0) &= 0.25 - 2\xi_1 + \xi_2; \\ P(X=1) &= 0.5 - 2\xi_2; \\ P(X=2) &= 0.25 + 2\xi_1 + \xi_2. \end{aligned}$$

Let $\hat{p}_0, \hat{p}_1, \hat{p}_2$ be sample proportions of $X = 0, 1, 2$. The likelihood would be maximized by setting

$$\begin{aligned} \hat{p}_0 &= 0.25 - 2\xi_1 + \xi_2; \\ \hat{p}_1 &= 0.5 - 2\xi_2; \\ \hat{p}_2 &= 0.25 + 2\xi_1 + \xi_2. \end{aligned}$$

The unconstrained solution is given by

$$\begin{aligned} \tilde{\xi}_1 &= (\hat{p}_2 - \hat{p}_0)/4 \\ \tilde{\xi}_2 &= (\hat{p}_2 + \hat{p}_0 - 0.5)/2 \end{aligned}$$

At the same time, because

$$\pi = 4\xi_1^2/\xi_2, \;\; \theta = 0.5(1 + \xi_2/\xi_1)$$

and they have range $[-0.25, 0.25] \times [0, 1]$, we must have

$$|\xi_2| \leq \xi_1, 4\xi_1^2 \leq \xi_2.$$

The range is shown in the following figure. In addition, after the shaded region is expanded, it is well approximated by a cone as show by the plot on the right hand side:

$$|\xi_2| \leq \xi_1, \xi_2 \geq 0.$$

Let φ be the angle from the positive x-axis. Then the cone contains two disjoint regions: $0 < \varphi < \pi/4$ and $3\pi/4 < \varphi < \pi/2$. This π is the mathematical constant for the ratio bewteen a circle's circumference to its diameter.

At the same time, under the null model, and using the classical central limit theorem,

$$\sqrt{n}(\tilde{\xi}_1, \tilde{\xi}_2) \xrightarrow{d} (Z_1, Z_2)$$

where (Z_1, Z_2) are bivariate normal with covariance matrix I^{-1} and I is the Fisher information with respect to (ξ_1, ξ_2):

$$I = \begin{bmatrix} 32 & 0 \\ 0 & 16 \end{bmatrix}.$$

The asymptotic variance can also be directly computed.

Applying Chernoff (1954), we may now regard the hypothesis test problem as testing

$$H_0 : \xi_1 = \xi_2 = 0$$

against the alternative

$$H_a : |\xi_2| \leq \xi_1, \xi_2 \geq 0$$

given a single pair of observation (Z_1, Z_2) with mean (ξ_1, ξ_2) and covariance matrix I^{-1} to obtain the limiting distribution of the original likelihood ratio test. Note that the log-likelihood function is given by

$$\ell(\xi_1, \xi_2) = -16(Z_1 - \xi_1)^2 - 8(Z_2 - \xi_2)^2 + c$$

where c is parameter free constant.

There are three cases depending on the observed value of (Z_1, Z_2) to obtain analytical solutions to the likelihood ratio test statistic.

Case I: $|Z_2| \leq Z_1, Z_2 \geq 0$. In this case the MLE $\hat{\xi}_1 = Z_1$ and $\hat{\xi}_2 = Z_2$. Hence, the likelihood ratio statistic is

$$T = -2\{\ell(\hat{\xi}_1, \hat{\xi}_2) - l(0,0)\} = 32Z_1^2 + 16Z_2^2 \sim \chi_2^2$$

Case II: $Z_2 < 0$. In this case the MLE $\hat{\xi}_1 = Z_1$ and $\hat{\xi}_2 = 0$. Hence, the likelihood ratio statistic is

$$T = -2\{\ell(\hat{\xi}_1, \hat{\xi}_2) - \ell(0,0)\} = 32Z_1^2 \sim \chi_1^2$$

Case II—: $|Z_1| > Z_2 \geq 0$. In this case, the likelihood is maximized when $\xi_1 = \xi_2$. the MLE $\hat{\xi}_1 = \hat{\xi}_2 = (2Z_1 + Z_2)/3$. Hence, the likelihood ratio statistic is

$$T = -2\{\ell(\hat{\xi}_1, \hat{\xi}_2) - \ell(0,0)\} = (16/3)(Z_2 - Z_1)^2$$

It can be verified that the event $\{|Z_2| \leq Z_1, Z_2 \geq 0\}$ in Case I is independent of $\{32Z_1^2 + 16Z_2^2 \leq z\}$ for any z. The same is true for the event in Case II. However, the event in the third case is not independent of $(Z_2 - Z_1)^2 \leq z$. Thus, we conclude the limiting distribution of the likelihood ratio test statistic T is given by

$$\begin{aligned} P(T \leq t) &= 0.5P(\chi_1^2 \leq t) + 2\lambda P(\chi_2^2 \leq t) \\ &\quad + (0.5 - 2\lambda)P\{(Z_2 - Z_1)^2 \leq 3t/16 || Z_1| > Z_2 \geq 0\} \end{aligned}$$

where $\lambda = \arctan(1/\sqrt{2})/(2\pi)$ and this $\pi = 3.14\cdots$.

Chernoff and Lander (1995) presented this result by introducing two standard normal Y_1, Y_2. It may not be obvious that this above result is the same.

Using the same approach, it is possible to go some distance. For instance, if one considers

$$H_0 : \pi(1-\pi)(\theta_2 - \theta_1) = 0$$

against the general alternative

$$H_0 : \pi(1-\pi)(\theta_2 - \theta_1) \neq 0.$$

Then if the null model is not $\theta_1 = \theta_2 = 0$, the limiting distribution of the likelihood ratio statistic when $m = 2$ is

$$0.5\chi_1^2 + 0.5\chi_2^2.$$

The limiting distribution when $m = 3$ can be similarly derived.

5.4 C(α) test

The general C(α) test is designed to test a specific null value of a parameter of interest in the presence of nuisance parameters. More specifically, suppose the statistical model is made of a family of density functions $f(x;\xi,\eta)$ with some appropriate parameter space for (ξ,η). The problem of interest is to test for $H_0 : \xi = \xi_0$ versus the alternative $H_a : \xi \neq \xi_0$. That is, the parameter value of η is left unspecified in both hypotheses and it is of no interest. This observation earns its name as a nuisance parameter. Due to its presence, the null and alternative hypotheses are composite as oppose to simple, as both contain more than a single parameter value in terms of (ξ,η). As in common practice of methodological development in statistical significance test, the size of the test is set at some α value between 0 and 1. Working on composite hypothesis and having size α appear to be the reasons behind the name C(α). While our interest lies in the use of C(α) to test for homogeneity in the context of mixture models, it is helpful to have a generic introduction.

5.4.1 The generic C(α) test

To motivate the C(α) test, let us first examine the situation where the model is free from nuisance parameters. Denote the model without nuisance parameter as a distribution family $f(x;\xi)$ with some parameter space in ξ. In addition, we assume that this family is regular. Namely, the density function is differentiable in ξ for all x, the derivative and the integration can be exchanged and so on. Based on an i.i.d. sample $x_1,\ldots,x_n$, the score function of ξ is given by

$$S_n(\xi) = \sum_{i=1}^{n} \frac{\partial \log f(x_i;\xi)}{\partial \xi}.$$

When the true distribution is given by $f(x;\xi_0)$, we have $\mathbb{E}\{S_n(\xi_0)\} = 0$. Define the Fisher information (matrix) to be

$$\mathbb{I}(\xi) = \mathbb{E}\left[\{\frac{\partial \log f(x_i;\xi)}{\partial \xi}\}\{\frac{\partial \log f(x_i;\xi)}{\partial \xi}\}^{\tau}\right].$$

It is well known that

$$S_n^{\tau}(\xi_0)\{n\mathbb{I}(\xi_0)\}^{-1}S_n(\xi_0) \xrightarrow{d} \chi_d^2$$

where d is the dimension of ξ. Clearly, a test for $H_0 : \xi = \xi_0$ versus the alternative $H_a : \xi \neq \xi_0$ based on S_n can be sensibly constructed with rejection region given by

$$S_n^{\tau}(\xi_0)\{\mathbb{I}^{-1}(\xi_0)\}S_n(\xi_0) \geq n\xi_d^2(1-\alpha).$$

We call it score test and credit its invention to Rao(??). When the dimension of ξ is $d = 1$, then the test can be equivalently defined based on the asymptotic normality of $S_n(\xi_0)$. In application, one may replace $n\mathbb{I}(\xi_0)$ by observed information and evaluate it at a root-n consistent estimator $\hat{\xi}$.

Back to the general situation where the model is given by $f(x;\xi,\eta)$. If η value in $f(x;\xi,\eta)$ is in fact known, the test problem reduces to the one we have just described. We may then proceed as follows. Define

$$S_n(\xi;\eta) = \sum_{i=1}^{n} \frac{\partial \log f(x_i;\xi,\eta)}{\partial \xi}.$$

The semicolon indicates that the "score" operation is with respect to only ξ. Similarly let us define the ξ-specific Fisher information to be

$$\mathbb{I}_{11}(\xi,\eta) = \mathbb{E}\left[\{\frac{\partial \log f(X;\xi,\eta)}{\partial \xi}\}\{\frac{\partial \log f(X;\xi,\eta)}{\partial \xi}\}^{\tau}\right].$$

With the value of η specified, we have a score test statistic and its limiting distribution

$$S_n^{\tau}(\xi_0;\eta)\{n\mathbb{I}_{11}(\xi_0,\eta)\}^{-1}S_n(\xi_0;\eta) \xrightarrow{d} \chi^2_d.$$

A score test can therefore be carried out using this statistic.

Without a known η value, the temptation is to have η replaced by an efficient or root-n consistent estimator. In general, the chisquare limiting distribution of

$$S_n^{\tau}(\xi_0;\hat{\eta})\{n\mathbb{I}(\xi_0;\hat{\eta})\}^{-1}S_n(\xi_0;\hat{\eta})$$

is no longer the simple chisquare. For a specific choice of $\hat{\eta}$, we may work out the limiting distribution of $S_n^{\tau}(\xi_0;\hat{\eta})$ and similarly define a new test statistic. The approach of Neyman (1959) achieved this goal in a graceful way.

To explain the scheme of Nayman, let us introduce the other score function

$$S_n(\eta;\xi) = \sum_{i=1}^{n} \frac{\partial \log f(x_i;\xi,\eta)}{\partial \eta}.$$

The above notation highlights that the "score operation" is with respect to only η. Next, let us define the other part of the Fisher information matrix to be

$$\mathbb{I}_{12}(\xi;\eta) = \mathbb{I}_{21}^{\tau}(\xi;\eta) = E\left[\{\frac{\partial \log f(X;\xi,\eta)}{\partial \xi}\}\{\frac{\partial \log f(X;\xi,\eta)}{\partial \eta}\}^{\tau}\right]$$

and

$$\mathbb{I}_{22}(\xi;\eta) = \mathbb{E}\left[\{\frac{\partial \log f(X;\xi,\eta)}{\partial \eta}\}\{\frac{\partial \log f(X;\xi,\eta)}{\partial \eta}\}^{\tau}\right].$$

Let us now project $S_n(\xi;\eta)$ into the orthogonal space of $S_n(\eta;\xi)$ to get

$$W_n(\xi,\eta) = S_n(\xi;\eta) - \mathbb{I}_{12}(\xi;\eta)\mathbb{I}_{22}^{-1}(\xi;\eta)S_n(\eta;\xi). \tag{5.5}$$

Conceptually, it removes the influence of the nuisance parameter η on the score function of ξ. At the true parameter value of ξ,η,

$$n^{-1/2}W_n(\xi,\eta) \xrightarrow{d} N(0,\{\mathbb{I}_{11} - \mathbb{I}_{12}\mathbb{I}_{22}^{-1}\mathbb{I}_{21}\}).$$

Under the null hypothesis, the value of ξ is specified as ξ_0, the value of η is unspecified. Thus, we naturally try to construct a test statistic based on $W_n(\xi_0, \hat{\eta})$ where $\hat{\eta}$ is some root-n estimator of η. For this purpose, we must know the distribution of $W_n(\xi_0, \hat{\eta})$. The following result of Neyman (1959) makes the answer to this question simple.

Theorem 5.1 *Suppose $x_1, \ldots, x_n$ is an i.i.d. sample from $f(x; \xi, \eta)$. Let $W_n(\xi, \eta)$ be defined as* (5.5) *together with other accompanying notations. Let $\hat{\eta}$ be a root-n consistent estimator of η when $\xi = \xi_0$. We have*

$$W_n(\xi_0, \eta) - W_n(\xi_0, \hat{\eta}) = o_p(n^{1/2})$$

as $n \to \infty$, under any distribution where $\xi = \xi_0$.

Due to the above theorem, the limiting distribution of $W_n(\xi_0, \hat{\eta})$ is the same as that of $W_n(\xi_0, \eta)$ with (ξ_0, η) being the true parameter values of the model that generated the data. Thus,

$$W_n^{\tau}(\xi_0, \hat{\eta})[n\{\mathbb{I}_{11} - \mathbb{I}_{12}\mathbb{I}_{22}^{-1}\mathbb{I}_{21}\}]^{-1} W_n(\xi_0, \hat{\eta})$$

may be used as the final C(α) test statistic. The information matrix in the above definition is evaluated at $\xi_0, \hat{\eta}$, and the rejection region can be decided based on its chisquare limiting distribution.

If we choose $\hat{\eta}$ as the constrained maximum likelihood estimator given $\xi = \xi_0$, we would have $W_n(\xi_0, \hat{\eta}) = S_n(\xi_0; \hat{\eta})$ in (5.5). The projected score function $S_n(\xi_0, \eta)$ is one of many possible zero-mean functions satisfying some regularity properties. Neyman(1959) called such class of functions Cramér functions. Each Cramér function can be projected to obtain a corresponding W_n and therefore a test statistic for $H_0 : \xi = \xi_0$. Within this class, the test based on score function $S_n(\xi_0, \eta)$ is optimal: having the highest asymptotic power against some local alternatives. In general, if the local alternative is of two-sided nature, the optimality based on the notion of "uniformly most powerful" cannot be achieved. The local optimality has to be justified in a more restricted sense.

5.4.2 C(α) test for homogeneity

As shown in the last subsection, the C(α) test is designed to test for a special null hypothesis in the presence of some nuisance parameters. The most convincing

example of its application is for homogeneity test. Recall that a mixture model is represented by its density function in the form of

$$f(x;G) = \int_{\Theta} f(x;\theta) dG(\theta).$$

Neyman and Scott (1966) regarded the variance of the mixing distribution Ψ as the parameter of interest, and the mean and other aspects of Ψ as nuisance parameters. In the simplest situation where $\Theta = \mathbb{R}$, we may rewrite the mixture density function as

$$\varphi(x;\theta,\sigma,\Psi) = \int_{\Theta} f(x;\theta+\sqrt{\sigma}\xi) d\Psi(\xi) \tag{5.6}$$

such that the standardized mixing distribution $\Psi(\cdot)$ has mean 0 and variance 1. The null hypothesis is $H_0 : \sigma = 0$. The rational of the choice of $\sqrt{\sigma}$ instead of σ in the above definition will be seen later. Both θ and the mixing distribution Ψ are nuisance parameters.

The partial derivative of $\varphi(x;\theta,\sigma,\Psi)$ with respect to σ is given by

$$\frac{\partial \varphi(x;\theta,\sigma,\Psi)}{\partial \sigma} = \frac{\int_{\Theta} \xi f'(x;\theta+\sqrt{\sigma}\xi) d\Psi(\xi)}{2\sqrt{\sigma}\int_{\Theta} f(x;\theta+\sqrt{\sigma}\xi) d\Psi(\xi)}.$$

At $\sigma = 0$ or let $\sigma \downarrow 0$, we find

$$\left.\frac{\partial \varphi(x;\theta,\sigma,\Psi)}{\partial \sigma}\right|_{\sigma\downarrow 0} = \frac{f''(x;\theta)}{2f(x;\theta)}.$$

This is the score function for σ based on a single observation. We may notice that the choice of $\sqrt{\sigma}$ gives us the non-degenerate score function. The partial derivative of $\varphi(x;\theta,\sigma,\Psi)$ with respect to θ is given by

$$\frac{\partial \varphi(x;\theta,\sigma,\Psi)}{\partial \theta} = \frac{\int_{\Theta} f'(x;\theta+\sqrt{\sigma}\xi) d\Psi(\xi)}{\int_{\Theta} f(x;\theta+\sqrt{\sigma}\xi) d\Psi(\xi)}$$

which leads to score function for θ based on a single observation as:

$$\frac{\partial \varphi(x;\theta,0,\Psi)}{\partial \theta} = \frac{f'(x;\theta)}{f(x;\theta)}.$$

Both of them are free from the mixing distribution Ψ.

Let us now define

$$y_i(\theta) = \frac{f'(x_i;\theta)}{f(x_i;\theta)}, \quad z_i(\theta) = \frac{f''(x_i;\theta)}{2f(x_i;\theta)} \tag{5.7}$$

with x_i's being i.i.d. observations from the mixture model. The score functions based on the entire sample are $\sum_{i=1}^n Z_i(\theta)$ and $\sum_{i=1}^n Y_i(\theta)$ for the mean and variance of G. Based on the principle of deriving test statistic W_n in the last subsection, we first project $z_i(\theta)$ into space of $y_i(\theta)$, and make use of the residual $w_i(\theta) = z_i(\theta) - \beta(\theta)y_i(\theta)$. The regression coefficient

$$\beta(\theta) = \frac{\mathbb{E}\{Y_1(\theta)Z_1(\theta)\}}{\mathbb{E}\{Y_1^2(\theta)\}}.$$

We capitalized Y and Z to indicate their status as random variables. The expectation is with respect to the homogeneous model $f(x;\theta)$.

When $\hat{\theta}$ is the maximum likelihood estimator of θ under the homogeneous model assumption $f(x,\theta)$, the C(α) statistic has a simple form:

$$W_n = \frac{\sum_{i=1}^n W_i(\hat{\theta})}{\sqrt{n\nu(\hat{\theta})}} = \frac{\sum_{i=1}^n Z_i(\hat{\theta})}{\sqrt{n\nu(\hat{\theta})}} \tag{5.8}$$

with $\nu(\theta) = \mathbb{E}\{W_1^2(\theta)\}$. Because the parameter of interest is univariate, we can skip the step of creating a quadratic form. Clearly, W_n has standard normal limiting distribution and the homogeneity null hypothesis is one-sided. At a given significance level α, we reject the homogeneity hypothesis H_0 when $W_n > z_\alpha$. This is the C(α) test for homogeneity.

In deriving the C(α) statistic, we assumed the parameter space $\Theta = \mathbb{R}$. With this parameter space, if $G(\cdot)$ is a mixing distribution on Θ, so is $G((\theta - \theta^*)/\sigma)$ for any θ^* and $\sigma \geq 0$. We have made use of this fact in (5.5). If $\Theta = \mathbb{R}^+$ as in the Poisson mixture model where $\theta \geq 0$, $G((\theta - \theta^*)/\sigma)$ cannot be regarded as a legitimate mixing distribution for some θ^* and σ. In the specific example of Poisson mixture, one may re-parameterize model with $\xi = \log\theta$. However, there seems to be no unified approach in general, and the optimality consideration is at stake.

Whether or not the mathematical derivation of W_n can be carried out as we did earlier for other forms of Θ, the statistic W_n remains a useful metric on the plausibility of the homogeneity hypothesis. The limiting distribution of W_n remains the same and it is usefulness in detecting the population heterogeneity.

5.4.3 C(α) statistic under NEF-QVF

Many commonly used distributions in statistics belong to a group of natural exponential families with quadratic variance function (NEF-QVF; Morris, 1982). The examples include normal, Poisson, binomial, and exponential. The density function in one-parameter natural exponential family has a unified analytical form

$$f(x;\theta) = h(x)\exp\{x\phi - A(\phi)\},$$

with respect to some σ-finite measure, where $\theta = A'(\phi)$ represents the mean parameter. Let $\sigma^2 = A''(\phi)$ be the variance under $f(x;\theta)$. To be a member of NEF-QVF, there must exist constants a, b, and c such that

$$\sigma^2 = A''(\phi) = aA'(\phi) + bA'(\phi) + c = a\theta^2 + b\theta + c. \tag{5.9}$$

Namely, the variance is a quadratic function of the mean.

When the kernel density function $f(x;\theta)$ is a member of NEF-QVF, the C(α) statistic has a particularly simple analytical form and simple interpretation.

Theorem 5.2 *When the kernel $f(x;\theta)$ is a member of NEF-QVF, then the*

$$W_n = \frac{\sum_{i=1}^n (x_i - \bar{x})^2 - n\hat{\sigma}^2}{\sqrt{2n(a+1)}\hat{\sigma}^2},$$

where C(α) statistic is given by $\bar{x} = n^{-1}\sum_{i=1}^n x_i$ and $\hat{\sigma}^2 = a\bar{x}^2 + b\bar{x} + c$ with coefficients given by (5.9) *are the maximum likelihood estimators of θ and σ^2, respectively.*

The analytical form of the C(α) test statistics for the normal, Poisson, binomial, and exponential kernels are included in Table 5.4.3 for easy reference. Their derivation is given in the next subsection.

Table 5.1: Analytical form of C(α) some NEF-QVF mixtures.

Kernel	$N(\theta,1)$	Poisson(θ)	BIN(m,p)	Exp(θ)
C(α)	$\frac{\sum_{i=1}^n (x_i-\bar{x})^2 - n}{\sqrt{2n}}$	$\frac{\sum_{i=1}^n (x_i-\bar{x})^2 - n\bar{x}}{\sqrt{2n\bar{x}}}$	$\frac{\sum_{i=1}^n (x_i-\bar{x})^2 - n\bar{x}(m-\bar{x})/m}{\sqrt{2n(1-1/m)\bar{x}(m-\bar{x})/m}}$	$\frac{\sum_{i=1}^n (x_i-\bar{x})^2 - n\bar{x}^2}{\sqrt{4n\bar{x}^2}}$

Note that the C(α) statistics contains the factor $\sum_{i=1}^{n}(x_i-\bar{x})^2$ which is a scaled up sample variance. The second term in the numerator of these C(α) statistics is the corresponding 'estimated variance' if the data are from the corresponding homogeneous NEF-QVF distribution. Hence, in each case, the test statistic is the difference between the 'observed variance' and the 'perceived variance' when the null hypothesis is true under the corresponding NEF-QVF kernel distribution assumption. The difference is then divided by their estimated null asymptotic variance. Thus, the C(α) test is the same as the 'over-dispersion' test.

5.4.4 Expressions of the C(α) statistics for NEF-VEF mixtures

The quadratic variance function under the natural exponential family is characterized by its density function $f(x;\theta)=h(x)\exp\{x\phi-A(\phi)\}$ and $A''(\phi)=aA'(\phi)+bA'(\phi)+c$ for some constants a, b, and c. The mean and variance are given by $\theta=A'(\phi)$ and $\sigma^2=A''(\phi)$. Taking derivatives with respect to ϕ on the quadratic relationship, we find that

$$\begin{aligned} A'''(\phi) &= \{2aA'(\phi)+b\}A''(\phi)=(2a\theta+b)\sigma^2, \\ A^{(4)}(\phi) &= 2a\{A''(\phi)\}^2+\{2aA'(\phi)+b\}A'''(\phi)=2a\sigma^4+(2a\theta+b)^2\sigma^2. \end{aligned}$$

Because of the regularity of the exponential family, we have

$$\mathscr{C}\left\{\frac{d^k f(X;\theta^*)/d\phi^k}{f(X;\theta^*)}\right\}=0$$

for $k=1,\ 2,\ 3,\ 4$ where θ^* is the true parameter value under the null model. This implies

$$\begin{aligned} \mathscr{C}\{(X-\theta^*)^3\} &= A'''(\phi^*)=(2a\theta^*+b)\sigma^{*2}, \\ \mathscr{C}\{(X-\theta^*)^4\} &= 3\{A''(\phi^*)\}^2+A^{(4)}(\phi^*)=(2a+3)\sigma^{*4}+(2a\theta^*+b)^2\sigma^{*2}, \end{aligned}$$

where ϕ^* is the value of the natural parameter corresponding to θ^*, and similarly for σ^{*2}.

The ingredients of the C(α) statistics are

$$\begin{aligned} Y_i(\theta^*) &= \frac{f'(X;\theta^*)}{f(X_i;\theta^*)}=\frac{(X_i-\theta^*)}{\sigma^{*2}}, \\ Z_i(\theta^*) &= \frac{f''(X;\theta^*)}{2f(X_i;\theta^*)}=\frac{(X_i-\theta^*)^2-(2a\theta^*+b)(X_i-\theta^*)-\sigma^{*2}}{2\{\sigma=4}. \end{aligned}$$

We then have

$$\mathscr{C}\{Y_i(\theta^*)Z_i(\theta^*)\} = \frac{\mathscr{C}\{(X_i-\theta^*)^3\} - (2a\theta^*+b)\mathscr{C}\{(X_i-\theta^*)^2\} - \sigma^{*2}\mathscr{C}(X_i-\theta^*)}{2\sigma^{*6}} = 0.$$

Therefore, the regression coefficient of $Z_i(\theta^*)$ against $Y_i(\theta^*)$ is $\beta(\theta^*) = 0$. This leads to the projection

$$W_i(\theta^*) = Z_i(\theta^*) - \beta(\theta^*)Y_i(\beta^*) = Z_i(\theta^*)$$

and

$$\begin{aligned} 4\{\sigma^*\}^8 \text{VAR}\{W_i(\theta^*)\} &= \text{VAR}\{(X_i-\theta^*)^2\} - 2(2a\theta^*+b)\mathscr{C}\{(X_i-\theta^*)^3\} \\ &\quad +(2a\theta^*+b)^2\mathscr{C}\{(X_i-\theta^*)^2\} \\ &= (2a+3)\{\sigma^*\}^4 + (2a\theta^*+b)^2\{\sigma^*\}^2 - \{\sigma^*\}^4 \\ &\quad -2(2a\theta^*+b)^2\{\sigma^*\}^2 + (2a\theta^*+b)^2\{\sigma^*\}^2 \\ &= (2a+2)\{\sigma^*\}^4. \end{aligned}$$

Hence, $\nu(\theta^*) = \text{VAR}\{W_i(\theta^*)\} = 0.5(a+1)\{\sigma^*\}^{-4}$.

Because the maximum likelihood estimator $\hat{\theta} = \bar{X}$, we find that

$$\sum_{i=1}^n W_i(\hat{\theta}) = \sum_{i=1}^n Z_i(\hat{\theta}) = \frac{\sum_{i=1}^n (X_i-\bar{X})^2 - \hat{\sigma}^2}{2\hat{\sigma}^4}$$

with $\hat{\sigma}^2 = a\bar{X}^2 + b\bar{X} + c$ due to invariance. The C(α) test statistic, $T_n = \sum_{i=1}^n W_i(\hat{\theta})/\sqrt{n\nu(\hat{\theta})}$, is therefore given by the simplified expression in Theorem 5.2. $\diamondsuit$

5.5 Brute-force likelihood ratio test for homogeneity

While C(α) is largely successful at homogeneity test, statisticians remain faithful to the likelihood ratio test for homogeneity. We have already given the example of Hartigan (1985) which shows the straightforward likelihood ratio statistic has stochastically unbounded. If we insists on the use of likelihood ratio test for homogeneity in general, we must either re-scale the statistics, or confine the model space so that the test statistic has a non-degenerative limiting distribution. The result of Chernoff and Lander (1995) on binomial mixture is the example where the

parameter space is naturally bounded. The result of Bickel and Chernoff (1993) is on re-scaled likelihood ratio statistic. The more general results are not easy to obtain. The first such attempt is by Basu and Ghosh (1995) who obtained the limiting distribution of the likelihood ratio statistic under a separate condition. Their result is a breakthrough in one way, but is short of providing the satisfactory answer to the limiting distribution.

In this section, we work on the original likelihood ratio statistics for homogeneity test with sufficient generality yet still limited in terms of developing a useful inference tool.

Let $f(x;\theta)$ for $\theta \in \Theta \subset \mathbb{R}$ be a p.d.f. with respect to a σ-finite measure. We observe a random sample $X_1,\ldots,X_n$ of size n from a population with the mixture p.d.f.

$$f(x;G) = (1-\pi)f(x;\theta_1) + \pi f(x;\theta_2), \tag{5.10}$$

where $\theta_j \in \Theta$, $j = 1,2$ are component parameter values and π and $1-\pi$ are the mixing proportions. Without loss of generality, we assume $0 \le \pi \le 1/2$. In this set up, the mixing distribution G has at most two distinct support points θ_1 and θ_2. We wish to test

$$H_0 : \alpha = 0 \text{ or } \theta_1 = \theta_2,$$

versus the full model (5.10).

As usual, the log likelihood function of the mixing distribution is given by

$$\ell_n(G) = \sum_{i=1}^{n} \log\{(1-\pi)f(x_i;\theta_1) + \pi f(x_i;\theta_2)\}.$$

The maximum likelihood estimator of G is a mixing distribution on Θ with at most two distinct support points at which $\ell_n(G)$ attains it maximum value. Without loss of generality, $\hat{\pi} < 1/2$.

We assume the finite mixture model (5.10) satisfies conditions for the consistency of $\hat{G}$. The consistency of $\hat{G}$ is of particular interest when the true mixing distribution G^* degenerates. A detailed analysis leads to the following results.

Lemma 5.1 *Suppose $\hat{G}$ is consistent as discussed in this section. As $n \to \infty$, both the MLE's of $\theta_1 - \theta^*$ and $\pi(\theta_2 - \theta^*)$ converge to 0 in probability when $f(x;\theta^*)$ is the true distribution.*

Proof: Let us denote the MLE of G as

$$\hat{G}(\theta) = (1-\hat{\pi})I(\hat{\theta}_1 \leq \theta) + \hat{\pi} I(\hat{\theta}_2 \leq \theta)$$

with $\hat{\pi} \leq 1/2$. The true mixing distribution under homogeneity model is $G^*(\theta) = I(\theta^* \leq \theta)$. Since Θ is compact, we find $\delta = \inf_{\theta \in \Theta} \exp(-|\theta|) > 0$. Thus, for the distance defined by (**??**), we have

$$\begin{aligned} D(\hat{G}, G^*) &= \int_\Theta |\hat{G}(\theta) - G^*(\theta)| \exp(-|\theta|) d\theta & (5.11) \\ &\geq \delta \int_\Theta |\hat{G}(\theta) - G^*(\theta)| d\theta & (5.12) \\ &= \delta\{(1-\hat{\pi})|\hat{\theta}_1 - \theta^*| + \hat{\pi}|\hat{\theta}_2 - \theta^*|\}. & (5.13) \end{aligned}$$

Since $\hat{\pi} < 1/2$, the consistency of $\hat{G}$ implies both $|\hat{\theta}_1 - \theta^*| \to 0$ and $\hat{\pi}|\hat{\theta}_2 - \theta^*| \to 0$ almost surely. These are the conclusions of this lemma. $\diamondsuit$.

The likelihood ratio test statistic is twice of the differences between two maximum values. We have discussed the one under the alternative model. When confined in the space of null hypothesis, the log likelihood function is simplified to

$$\ell_n(\theta) = \sum_{i=1}^n \log\{f(x_i; \theta)\}.$$

Note that as usual, $\ell_n(\cdot)$ has been used as a generic notation. Denote its global maximum point in Θ as $\hat{\theta}$. The likelihood ratio test statistic is then

$$R_n = 2\{\ell_n(\hat{G}) - \ell_n(\hat{\theta})\}.$$

We now focus on deriving its limiting distribution under a few additional conditions.

Strong identifiability. The kernel function $f(x;\theta)$ together with the first two derivatives $f'(x;\theta)$ and $f''(x;\theta)$ are identifiable by θ. That is, for any $\theta_1 \neq \theta_2$ in Θ,

$$\sum_{j=1}^2 \{a_j f(x, \theta_j) + b_j f'(x, \theta_j) + c_j f''(x, \theta_j)\} = 0,$$

for all x implies that $a_j = b_j = c_j = 0$, $j = 1, 2$.

The identifiability required here is stronger than ordinary in the sense that, besides $f(x,\theta)$ itself, the first two derivatives are also identifiable. This was first proposed by Chen (1995) to establish that the best possible rate of estimating G is $n^{-1/4}$ under some condition. This topic will be discussed further in another chapter.

The following quantities play important roles in our study:

$$Y_i(\theta) = \frac{f(X_i,\theta)-f(X_i,\theta^*)}{(\theta-\theta^*)f(X_i,\theta^*)}; \tag{5.14}$$

$$Z_i(\theta) = \frac{Y_i(\theta)-Y_i(\theta^*)}{\theta-\theta^*}. \tag{5.15}$$

At $\theta=\theta^*$, the above functions taking continuity limits as their values. The projection residual of Z to the space of Y is denoted as

$$W_i(\theta) = Z_i(\theta) - h(\theta)Y_i(\theta^*), \tag{5.16}$$

where $h(\theta) = E\{Y_i(\theta^*)Z_i(\theta)\}/E\{Y_i^2(\theta^*)\}$. These notations match the ones defined in the section for C(α) test with a minor change: $Y_i(\theta^*)$ differs by a factor of 2.

Both $Y_i(\theta)$ and $Z_i(\theta)$ are continuous with $E\{Y_i(\theta)\}=0$ and $E\{Z_i(\theta)\}=0$ and that $Z_i(\theta)$ can be approximated by the derivative of $Y_i(\theta)$ through the mean value theorem. They were regarded as score functions in the contents of C(α) test.

Uniform integrable upper bound. There exists integrable g with some $\delta>0$ such that $|Y_i(\theta)|^{4+\delta} \le g(X_i)$ and $|Y_i'(\theta)|^3 \le g(X_i)$ for all $\theta\in\Theta$.

Note that $Y_i(\theta)$ is uniformly continuous in X_i and θ over $S\times\Theta$, where S is any compact interval of real numbers. This implies equicontinuity of $Y_i(\theta)$ in θ for $X_i\in S$. According to Rubin (1956), this condition ensures that $n^{-1}\sum_{i=1}^n |Y_i(\theta)|^k$ converges almost surely to $E\{|Y_1(\theta)|^k\}$ uniformly in $\theta\in\Theta$, for $k\le 4$. The same results hold for $n^{-1}\sum_{i=1}^n |Z_i(\theta)|^k$.

Lemma 5.2 *Suppose the model (5.10) satisfies the strong identifiability condition and the uniform integrable upper bound conditions. Then the covariance matrix of the vector* $(Y_1(\theta^*),Z_1(\theta))'$ *is positive definite at all* $\theta\in\Theta$ *under the homogeneous model with* $\theta=\theta^*$.

Proof. The result is implied by Cauchy inequality

$$\mathscr{C}^2\{Y_1(\theta^*)Z_1(\theta)\} \leq \mathscr{C}\{Y_1^2(\theta^*)\}\mathscr{C}\{Z_1^2(\theta)\},$$

where the equality holds only if $Y_1(\theta^*)$ and $Z_1(\theta)$ are linearly associated. The strong identifiability condition thus ensures that the inequality holds. The integrable upper bound conditions makes these expectations exist. ♢

The limiting distribution of the likelihood ratio statistic R_n will be described by some stochastic process. The convergence of a sequence of stochastic process is an indispensable notion.

Lemma 5.3 *The processes $n^{-1/2}\sum Y_i(\theta)$, $n^{-1/2}\sum Y_i'(\theta)$, $n^{-1/2}\sum Z_i(\theta)$ and $n^{-1/2}\sum Z_i'(\theta)$ over Θ are tight.*

Proof. To see this, consider

$$\begin{aligned} E\left\{n^{-1/2}\sum_{i=1}^{n} Y_i(\theta_2) - n^{-1/2}\sum_{i=1}^{n} Y_i(\theta_1)\right\}^2 &= E\{Y_1(\theta_2) - Y_1(\theta_1)\}^2 \\ &\leq E\left\{g^{2/3}(X_1)(\theta_2-\theta_1)^2\right\}. \end{aligned}$$

By Theorem 12.3 of Billingsley (1968, p. 95), $n^{-1/2}\sum Y_i(\theta)$ is tight. From this comment, we know that a sufficient condition for the tightness of $n^{-1/2}\sum Y_i'(\theta)$ and $n^{-1/2}\sum Z_i'(\theta)$ is that $\{Y_i''(\theta)\}^2 \leq g(X_i)$ and $\{Z_i''(\theta)\}^2 \leq g(X_i)$ with an integrable function g. ♢

Theorem 5.3 *Suppose the mixture model (5.10) satisfies all conditions specified in this section. When $f(x,\theta^*)$ is the true null distribution, the asymptotic distribution of the likelihood ratio test statistic for homogeneity is that of*

$$\{\sup_{\theta\in\Theta} W^+(\theta)\}^2,$$

where $W(\theta)$, $\theta \in \Theta$, is a Gaussian process with mean 0, variance 1 and autocorrelation function $\rho(\theta,\theta')$ given by

$$\rho(\theta,\theta') = \frac{\text{cov}\{W_1(\theta), W_1(\theta')\}}{\sqrt{E\{W_1^2(\theta)\}E\{W_1^2(\theta')\}}}, \tag{5.17}$$

and that $W_1(\theta)$ is defined by (5.16).

This result is first presented in Chen and Chen (199?). We notice that the result of ?? is more general. If one works hard enough, this result can be directly obtained from that one. However, the result here is presented in a much more comprehensive fashion.

5.5.1 Examples

The autocorrelation function $\rho(\theta,\theta')$ of the Gaussian process $W(\theta)$ is well behaved with commonly-used kernel functions. In particular, when $Z_1(\theta)$ and $Y_1(\theta^*)$ are uncorrelated, i.e., $h(\theta) = 0$, $\rho(\theta,\theta')$ becomes simple. We provide its expressions for normal, binomial and Poisson. As shown in the last section, $Z_1(\theta)$ and $Y_1(\theta^*)$ are uncorrelated. In terms of Y_i and Z_i,

$$\rho(\theta,\theta') = \frac{\text{COV}\{Z_1(\theta) - h(\theta)Y_1(\theta^*), Z_1(\theta') - h(\theta')Y_1(\theta^*)\}}{\sqrt{\text{VAR}\{Z_1(\theta) - h(\theta)Y_1(\theta^*)\}\text{VAR}\{Z_1(\theta') - h(\theta')Y_1(\theta^*)\}}},$$

where $h(\theta) = E\{Z_1(\theta)Y_1(\theta^*)\}/E\{Y_1^2(\theta^*)\}$. When $Z_1(\theta)$ and $Y_1(\theta^*)$ are uncorrelated,

$$\rho(\theta,\theta') = \frac{\text{COV}\{Z_1(\theta), Z_1(\theta')\}}{\sqrt{\text{VAR}\{Z_1(\theta)\}\text{VAR}\{Z_1(\theta')\}}}.$$

Example 5.1 Normal kernel function. *Let $f(x;\theta)$ be normal $N(\theta,\sigma^{*2})$ with known σ^*. For simplicity, let $\theta^* = 0$ and $\sigma^* = 1$. Then $Y_1(0) = X_1$ and for $\theta \neq 0$,*

$$Z_1(\theta) = \theta^{-1}\{\theta^{-1}(\exp\{X_1\theta - \theta^2/2\} - 1) - X_1\},$$

and $Z_1(0) = (X_1^2 - 1)/2$. We have seen that $E\{Z_1(\theta)Y_1(0)\} = 0$, and hence $h(\theta) = 0$ for all θ. Note that for θ, $\theta' \neq 0$,

$$E\{Z_1(\theta)Z_1(\theta')\} = (\theta\theta')^{-2}\{\exp(\theta\theta') - 1 - \theta\theta'\}.$$

We have for $\theta,\ \theta' \neq 0$,

$$\rho(\theta,\theta') = \frac{\exp(\theta\theta') - 1 - \theta\theta'}{\sqrt{\{\exp(\theta^2) - 1 - \theta^2\}\{\exp(\theta'^2) - 1 - \theta'^2\}}}.$$

For $\theta \neq 0$, *it reduces to*

$$\rho(\theta,0) = \frac{\theta^2}{\sqrt{2\{\exp(\theta^2) - 1 - \theta^2\}}}.$$

♢

Example 5.2 Binomial kernel function. *Consider the binomial kernel function*

$$f(x,\theta) \propto \theta^x(1-\theta)^{k-x}, \textit{for } x = 0,\ldots,k.$$

Note that

$$Y_1(\theta^*) = \frac{X_1}{\theta^*} - \frac{k - X_1}{1-\theta^*}; \tag{5.18}$$

$$Y_1(\theta) = \frac{1}{\theta - \theta^*}\left[\left(\frac{1-\theta}{1-\theta^*}\right)^k \left\{\frac{(1-\theta^*)/\theta^*}{(1-\theta)/\theta}\right\}^{X_1} - 1\right]. \tag{5.19}$$

As a member of NEF-QVF, binomial mixture models also have $E\{Z_1(\theta)Y_1(\theta^*)\} = 0$, *so* $h(\theta) = 0$ *for all* θ. *The covariance of* $Z_1(\theta)$ *and* $Z_1(\theta')$ *for* $\theta, \theta' \neq \theta^*$ *is*

$$\frac{1}{(\theta-\theta^*)^2(\theta'-\theta^*)^2}\left[\left\{1 + \frac{(\theta-\theta^*)(\theta'-\theta^*)}{\theta^*(1-\theta^*)}\right\}^k - 1 - k\frac{(\theta-\theta^*)(\theta'-\theta^*)}{\theta^*(1-\theta^*)}\right].$$

Let

$$u = \sqrt{\frac{k(\theta-\theta^*)}{\theta^*(1-\theta^*)}}, \quad u' = \sqrt{\frac{k(\theta'-\theta^*)}{\theta^*(1-\theta^*)}}.$$

We obtain

$$\rho(\theta,\theta') = \frac{(1+uu'/k)^k - 1 - uu'}{\sqrt{\left\{(1+u^2/k)^k - 1 - u^2\right\}\left\{(1+u'^2/k)^k - 1 - u'^2\right\}}}.$$

Interestingly, when k is large,

$$\rho(\theta,\theta') \approx \frac{\exp(uu') - 1 - uu'}{\sqrt{\{\exp(u^2) - 1 - u^2\}\{\exp(u'^2) - 1 - u'^2\}}}.$$

That is, when k is large the autocorrelation function behaves similarly to that of the normal kernel. ♢

Example 5.3 Poisson kernel function. *Let*

$$f(x,\theta) \propto e^{-\theta}\theta^x, \textit{for } x = 0,1,2,\ldots.$$

Then

$$Y_1(\theta^*) = \frac{X_1 - \theta^*}{\theta^*}; \tag{5.20}$$

$$Y_1(\theta) = \frac{\exp\{-(\theta-\theta^*)\}(\theta/\theta^*)^{X_1} - 1}{\theta - \theta^*} \textit{ for } \theta \neq \theta^*. \tag{5.21}$$

Again Poisson is a member of NEF-QVF so that $Z_1(\theta)$ and $Y_1(\theta^*)$ are uncorrelated and $h(\theta) = 0$ for all θ.

For $\theta, \theta' \neq \theta^*$,

$$\text{COV}\{Z_1(\theta), Z_1(\theta')\} = \frac{\exp\{(\theta-\theta^*)(\theta'-\theta^*)/\theta^*\} - 1 - (\theta'-\theta^*)/\theta^*}{(\theta-\theta^*)^2(\theta'-\theta^*)^2}.$$

Put

$$v = \frac{\theta-\theta^*}{\sqrt{\theta^*}}, \quad v' = \frac{\theta'-\theta^*}{\sqrt{\theta^*}}.$$

Then

$$\rho(\theta,\theta') = \frac{(1+vv'/k)^k - 1 - vv'}{\sqrt{\left\{(1+v^2/k)^k - 1 - v^2\right\}\left\{(1+v'^2/k)^k - 1 - v'^2\right\}}}.$$

Interestingly, this form is identical to the one for normal kernel. ♢

We can easily verify all conditions of the theorem are satisfied in these examples, when the parameter space Θ is confined to a compact subset of $\mathbb{R}$.

Exponential distribution is another popular member of NEF-QVF. This distribution does not satisfy the integrable upper bound condition in general. It is somewhat a surprise that many of results developed in the literature are not applicable to the exponential mixture model.

5.5.2 The proof of Theorem 5.2

We carefully analyze the asymptotic behaviour of the likelihood ratio test statistics under the true distribution $f(x;\theta^*)$. For convenience, we rewrite the log likelihood function under two-component mixture model as

$$\ell_n(\pi,\theta_1,\theta_2)=\sum_{i=1}^{n}\log\{(1-\pi)f(X_i;\theta_1)+\pi f(X_i;\theta_2)\}.$$

The new notation helps to highlight the detailed structure of the mixing distribution G. As usual, we have used $\ell_n(\cdot)$ in a very generic and non-rigorous fashion. Define

$$r_n(\pi,\theta_1,\theta_2)=2\{\ell_n(\pi,\theta_1,\theta_2)-\sup_{\theta\in\Theta}\ell_n(0,\theta,\theta)\}$$

and

$$R_n=\sup\{r_n(\pi,\theta_1,\theta_2):\pi,\theta_1,\theta_2\}$$

over the region of $0\le\pi\le 1/2$, $\theta_j\in\Theta$, $j=1,2$. We may call $r_n(\pi,\theta_1,\theta_2)$ log likelihood ratio functions.

To find the asymptotic distribution of R_n, it is convenient to partition r_n into two parts:

$$\begin{aligned} r_n(\pi,\theta_1,\theta_2) &= 2\{\ell_n(\pi,\theta_1,\theta_2)-\ell_n(0,\theta^*,\theta^*)\}+2\{\ell_n(0,\theta^*,\theta^*)-\ell_n(0,\hat\theta,\hat\theta)\} \\ &= r_{1n}(\pi,\theta_1,\theta_2)+r_{2n}, \end{aligned}$$

where $\hat\theta$ is the MLE of θ under the null model. Note that

$$-r_{2n}=-2\{\ell_n(0,\theta^*,0)-\ell_n(0,\hat\theta,\hat\theta)\}$$

is an ordinary likelihood ratio (no mixtures involved) and hence an approximation is immediate as follows (Wilks 1938):

$$r_{2n}=-\frac{\{n^{-1/2}\sum_{i=1}^{n}Y_i(\theta^*)\}^2}{E\{Y_1^2(\theta^*)\}}+o_p(1). \tag{5.22}$$

This expansion suffices in deriving the limiting distribution of R_n. The main task is to analyze $r_{1n}(\pi,\theta_1,\theta_2)$.

Write

$$r_{1n}(\pi,\theta_1,\theta_2)=2\sum_{i=1}^{n}\log(1+\delta_i),$$

where

$$\begin{aligned}\delta_i &= (1-\pi)\left\{\frac{f(X_i,\theta_1)}{f(X_i,\theta^*)}-1\right\}+\pi\left\{\frac{f(X_i,\theta_2)}{f(X_i,\theta^*)}-1\right\}\\ &= (1-\pi)(\theta_1-\theta^*)Y_i(\theta_1)+\pi(\theta_2-\theta^*)Y_i(\theta_2) \qquad (5.23)\end{aligned}$$

and $Y_i(\theta)$ is defined by (5.14).

The main idea is as follows. By the Taylor expansion,

$$r_{1n}(\pi,\theta_1,\theta_2)=2\sum_{i=1}^{n}\delta_i-\sum_{i=1}^{n}\delta_i^2+\varepsilon_n.$$

We need to argue that when the sample size n is large, the remainder ε_n is negligible uniformly in the mixing parameters. Negligibility relies on consistency of the MLE's of the parameters. By Lemma 5.1, the MLE of θ_1 is consistent. The problem is that the MLE's of θ_2 and π

Our solution is to consider the case of $|\theta_2-\theta^*|\le\varepsilon$ for an arbitrarily small $\varepsilon>0$, and that of $|\theta_2-\theta^*|>\varepsilon$, separately. In the process, we used sandwich idea. Let $R_n(\varepsilon;I)$ denote the supremum of $r_n(\pi,\theta_1,\theta_2)$ with the restriction $|\theta_2-\theta^*|>\varepsilon$, and $R_n(\varepsilon;II)$ the supremum with $|\theta_2-\theta^*|\le\varepsilon$.

Case I: $|\theta_2-\theta^*|>\varepsilon$.

In this case, let $\hat\pi(\theta_2)$ and $\hat\theta_1(\theta_2)$ be the MLE's of π and θ_1 with fixed $\theta_2\in\Theta$ such that $|\theta_2-\theta^*|>\varepsilon$. The consistency results of Lemma 5.1 remain true for $\hat\pi(\theta_2)$ and $\hat\theta_1(\theta_2)$. For simplicity of notation, we write $\hat\pi=\hat\pi(\theta_2)$ and $\hat\theta_1=\hat\theta_1(\theta_2)$.

First we establish an upper bound on $R_n(\varepsilon;I)$. By the inequality $2\log(1+x)\le 2x-x^2+(2/3)x^3$, we have

$$r_{1n}(\pi,\theta_1,\theta_2)=2\sum_{i=1}^{n}\log(1+\delta_i)\le 2\sum_{i=1}^{n}\delta_i-\sum_{i=1}^{n}\delta_i^2+\frac{2}{3}\sum_{i=1}^{n}\delta_i^3, \qquad (5.24)$$

where

$$\delta_i=(1-\pi)(\theta_1-\theta^*)Y_i(\theta_1)+\pi(\theta_2-\theta^*)Y_i(\theta_2)$$

as defined in (5.23). Replacing θ_1 with θ^* in the quantity $Y_i(\theta_1)$ gives

$$\delta_i=(1-\pi)(\theta_1-\theta^*)Y_i(\theta^*)+\pi(\theta_2-\theta^*)Y_i(\theta_2)+e_{in},$$

where the remainder is given by

$$e_{in} = (1-\pi)(\theta_1 - \theta^*)\{Y_i(\theta_1) - Y_i(\theta^*)\}.$$

In terms of Z_i defined in (**??**), $e_{in} = (1-\pi)(\theta_1-\theta^*)^2 Z_i(\theta_1)$. Since $n^{-1/2}\sum_{i=1}^n Z_i(\theta_1)$ is tight, it is $O_p(1)$, so

$$e_n = \sum_{i=1}^n e_{in} = n^{1/2}(1-\pi)(\theta_1-\theta^*)^2 n^{-1/2}\sum_{i=1}^n Z_i(\theta_1) = n^{1/2}(\theta_1-\theta^*)^2 O_p(1).$$

Similarly, replace $Y_i(\theta_1)$ with $Y_i(\theta^*)$ in the square and cubic terms of δ_i, which results in a remainder having a higher order than e_n. Consequently,

$$\begin{aligned} r_{1n}(\pi,\theta_1,\theta_2) \;\leq\; & 2\sum_{i=1}^n \{(1-\pi)(\theta_1-\theta^*)Y_i(\theta^*) + \pi(\theta_2-\theta^*)Y_i(\theta_2)\} \\ & - \sum_{i=1}^n \{(1-\pi)(\theta_1-\theta^*)Y_i(\theta^*) + \pi(\theta_2-\theta^*)Y_i(\theta_2)\}^2 \\ & + \frac{2}{3}\sum_{i=1}^n \{(1-\pi)(\theta_1-\theta^*)Y_i(\theta^*) + \pi(\theta_2-\theta^*)Y_i(\theta_2)\}^3 \\ & + n^{1/2}(\theta_1-\theta^*)^2 O_p(1). \end{aligned}$$

Now write

$$(1-\pi)(\theta_1-\theta^*)Y_i(\theta^*) + \pi(\theta_2-\theta^*)Y_i(\theta_2) = m_1 Y_i(\theta^*) + m_2 Z_i(\theta_2),$$

where

$$\begin{aligned} m_1 &= (1-\pi)(\theta_1-\theta^*) + \pi(\theta_2-\theta^*); && (5.25) \\ m_2 &= \pi(\theta_2-\theta^*)^2. && (5.26) \end{aligned}$$

Hence

$$\begin{aligned} r_{1n}(\pi,\theta_1,\theta_2) \;\leq\; & 2\sum_{i=1}^n \{m_1 Y_i(\theta^*) + m_2 Z_i(\theta_2)\} - \sum_{i=1}^n \{m_1 Y_i(\theta^*) + m_2 Z_i(\theta_2)\}^2 \\ & + \frac{2}{3}\sum_{i=1}^n \{m_1 Y_i(\theta^*) + m_2 Z_i(\theta_2)\}^3 + n^{1/2}(\theta_1-\theta^*)^2 O_p(1), \end{aligned} \quad (5.27)$$

Since

$$n^{-1}\sum\{|Y_i(\theta^*)|^3+|Z_i(\theta_2)|^3\}=O_p(1)$$

uniformly in θ_2, and since

$$n^{-1}\sum\{m_1Y_i(\theta^*)+m_2Z_i(\theta_2)\}^2$$

converges uniformly to a positive definite quadratic form in m_1 and m_2 uniformly in θ_2 (Lemma 5.2), it follows that

$$\frac{\sum|m_1Y_i(\theta^*)+m_2Z_i(\theta_2)|^3}{\sum\{m_1Y_i(\theta^*)+m_2Z_i(\theta_2)\}^2}\le\max(|m_1|,|m_2|)O_p(1).$$

Let

$$\hat{m}_1=(1-\hat{\pi})(\hat{\theta}_1-\theta^*)+\hat{\pi}(\theta_2-\theta^*),\quad \hat{m}_2=\hat{\pi}(\theta_2-\theta^*)^2.$$

By Lemma 1, $\max(|\hat{m}_1|,|\hat{m}_2|)=o_p(1)$ uniformly in θ_2. Consequently, it follows that

$$\sum_{i=1}^n\{\hat{m}_1Y_i(\theta^*)+\hat{m}_2Z_i(\theta_2)\}^3=o_p\left[\sum_{i=1}^n\{\hat{m}_1Y_i(\theta^*)+\hat{m}_2Z_i(\theta_2)\}^2\right]. \tag{5.28}$$

The remainder $n^{1/2}(\hat{\theta}_1-\theta^*)^2O_p(1)$ in (5.27) is also negligible when compared to the square terms. To see this, recall that $0\le\hat{\pi}\le 1/2$, $|\theta_2-\theta^*|\ge\varepsilon$ and $\hat{\theta}_1-\theta^*=o_p(1)$. We have

$$\begin{aligned}
n^{1/2}(\hat{\theta}_1-\theta^*)^2 &\le 4n^{1/2}\{|\hat{m}_1|+|\hat{\pi}(\theta_2-\theta^*)|\}^2\\
&\le 4n^{1/2}(|\hat{m}_1|+\hat{m}_2/\varepsilon)^2\\
&\le 8n^{1/2}(\hat{m}_1^2+\hat{m}_2^2/\varepsilon^2)\\
&= o_p\left[\sum_{i=1}^n\{\hat{m}_1Y_i(\theta^*)+\hat{m}_2Z_i(\theta_2)\}^2\right].
\end{aligned} \tag{5.29}$$

Combining (5.27), (5.28) and (5.29) yields

$$\begin{aligned}
r_{1n}(\hat{\pi},\hat{\theta}_1,\theta_2) \le\ & 2\sum_{i=1}^n\{\hat{m}_1Y_i(\theta^*)+\hat{m}_2Z_i(\theta_2)\}\\
&-\sum_{i=1}^n\{\hat{m}_1Y_i(\theta^*)+\hat{m}_2Z_i(\theta_2)\}^2\{1+o_p(1)\},
\end{aligned} \tag{5.30}$$

uniformly in θ_2.

To clarify the role each component plays, we can use the quantity $W_i(\theta)$ introduced in (5.16). Since $E\{Y_i(\theta^*)W_i(\theta)\} = 0$, $Y_i(\theta^*)$ and $W_i(\theta)$ are orthogonal so that

$$\sum_{i=1}^{n} Y_i(\theta^*)W_i(\theta_2) = O_p(n^{1/2}),$$

uniformly in θ_2. Hence (5.30) becomes

$$\begin{aligned} r_{1n}(\hat{\pi},\hat{\theta}_1,\theta_2) \quad &\leq \quad 2\sum_{i=1}^{n}\{\hat{t}Y_i(\theta^*) + \hat{m}_2 W_i(\theta_2)\} \\ &\quad - \{\hat{t}^2\sum_{i=1}^{n} Y_i^2(\theta^*) + \hat{m}_2^2\sum_{i=1}^{n} W_i^2(\theta_2)\}\{1+o_p(1)\}, \end{aligned}$$

where $\hat{t} = \hat{m}_1 + \hat{m}_2 h(\theta_2)$. For fixed θ_2, consider the quadratic function

$$Q(t,m_2) = 2\sum_{i=1}^{n}\{tY_i(\theta^*) + m_2 W_i(\theta_2)\} - \left\{t^2\sum_{i=1}^{n} Y_i^2(\theta^*) + m_2^2\sum_{i=1}^{n} W_i^2(\theta_2)\right\}. \tag{5.31}$$

Over the region of $m_2 \geq 0$, for any θ_2 fixed, $Q(t,m_2)$ is maximized at $t = \tilde{t}$ and $m_2 = \tilde{m}_2$, with

$$\tilde{t} = \frac{\sum Y_i(\theta^*)}{\sum Y_i^2(\theta^*)}, \quad \tilde{m}_2 = \frac{\{\sum W_i(\theta_2)\}^+}{\sum W_i^2(\theta_2)}. \tag{5.32}$$

Thus

$$r_{1n}(\hat{\pi},\hat{\theta}_1,\theta_2) \leq Q(\tilde{t},\tilde{m}_2) + o_p(1) = \frac{\{\sum_{i=1}^n Y_i(\theta^*)\}^2}{\sum_{i=1}^n Y_i^2(\theta^*)} + \frac{[\{\sum_{i=1}^n W_i(\theta_2)\}^+]^2}{\sum_{i=1}^n W_i^2(\theta_2)} + o_p(1).$$

Finally, by integrable upper bound condition,

$$n^{-1}\sum Y_i^2(\theta^*) = E\{Y_1^2(\theta^*)\} + o_p(1)$$

and

$$n^{-1}\sum W_i^2(\theta_2) = E\{W_1^2(\theta_2)\} + o_p(1).$$

Therefore,

$$r_{1n}(\hat{\pi},\hat{\theta}_1,\theta_2) \leq \frac{\{n^{-1/2}\sum_{i=1}^n Y_i(\theta^*)\}^2}{E\{Y_1^2(\theta^*)\}} + \frac{[\{n^{-1/2}\sum_{i=1}^n W_i(\theta_2)\}^+]^2}{E\{W_1^2(\theta_2)\}} + o_p(1). \tag{5.33}$$

From (5.33) and (5.22),

$$r_n(\hat{\pi},\hat{\theta}_1,\theta_2) = r_{1n}(\hat{\pi},\hat{\theta}_1,\theta_2) + r_{2n} \le \frac{[\{n^{-1/2}\sum W_i(\theta_2)\}^+]^2}{E\{W_1^2(\theta_2)\}} + o_p(1). \qquad (5.34)$$

Hence we have established an upper bound on $R_n(\varepsilon;I)$ as follows:

$$R_n(\varepsilon;I) \le \sup_{|\theta-\theta^*|>\varepsilon} \frac{[\{n^{-1/2}\sum_{i=1}^n W_i(\theta)\}^+]^2}{E\{W_1^2(\theta)\}} + o_p(1).$$

To obtain a lower bound of $R_n(\varepsilon;I)$, for any θ_2 fixed such that $|\theta_2-\theta^*| \ge \varepsilon$, let $\tilde{\pi}$ and $\tilde{\theta}_1$ to be the values determined by $\tilde{t}$ and $\tilde{m}_2$ as given in (5.32). Since $\tilde{t} = O_p(n^{-1/2})$, $\tilde{m}_2 = O_p(n^{-1/2})$ uniformly in θ_2, and $|h(\theta_2)| \le \sqrt{E\{Z_1^2(\theta_2)\}/E\{Y_1^2(\theta^*)\}}$ is a bounded quantity, it follows that $\tilde{\pi} = O_p(n^{-1/2})$ and $\tilde{\theta}_1 = O_p(n^{-1/2})$ uniformly in θ_2 satisfying $|\theta_2-\theta^*| \ge \varepsilon$. Consider the following Taylor expansion

$$r_{1n}(\tilde{\pi},\tilde{\theta}_1,\theta_2) = 2\sum_{i=1}^n \tilde{\delta}_i - \sum_{i=1}^n \tilde{\delta}_i^2(1+\tilde{\eta}_i)^{-2},$$

where $|\tilde{\eta}_i| < |\tilde{\delta}_i|$ and $\tilde{\delta}_i$ is the δ_i in (5.23) with $\pi = \tilde{\pi}$ and $\theta_1 = \tilde{\theta}_1$. We have

$$|\tilde{\delta}_i| \le (|\tilde{\theta}_1| + |\tilde{\pi}|M) \max_{1\le i\le n}[\sup_{\theta\in\Theta}\{|Y_i(\theta)|\}].$$

By integrable upper bound condition, $|Y_i(\theta)|^{4+\delta} \le g(X_i)$ and $E\{g(X_i)\}$ is finite. Hence

$$\max_{1\le i\le n}[\sup_{\theta\in\Theta}\{|Y_i(\theta)|\}] = o_p(n^{1/4}),$$

implying $\max(|\tilde{\eta}_i|) = o_p(1)$ uniformly. It follows that

$$r_{1n}(\tilde{\pi},\tilde{\theta}_1,\theta_2) = 2\sum_{i=1}^n \tilde{\delta}_i - \sum_{i=1}^n \tilde{\delta}_i^2\{1+o_p(1)\}.$$

Applying the argument leading to (5.34), we know that with $\tilde{\pi}$ and $\tilde{\theta}_1$,

$$r_n(\tilde{\pi},\tilde{\theta}_1,\theta_2) \ge \frac{[\{n^{-1/2}\sum_{i=1}^n W_i(\theta_2)\}^+]^2}{E\{W_1^2(\theta_2)\}} + o_p(1).$$

Therefore,

$$\sup_{\pi,\theta_1} r_n(\pi,\theta_1,\theta_2) \ge r_n(\tilde{\pi},\tilde{\theta}_1,\theta_2) \ge \frac{[\{n^{-1/2}\sum_{i=1}^n W_i(\theta_2)\}^+]^2}{E\{W_1^2(\theta_2)\}} + o_p(1).$$

Combining with (5.34), we thus arrive at the following result.

Lemma 5.4 *Suppose all conditions specified in this section. Let $R_n(\varepsilon;I)$ be the likelihood ratio statistic with restriction $|\theta_2-\theta^*|>\varepsilon$. Then when $f(x,\theta^*)$ is the true null distribution, as $n\to\infty$,*

$$R_n(\varepsilon;I)=\sup_{|\theta-\theta^*|>\varepsilon}\frac{[\{n^{-1/2}\sum_{i=1}^n W_i(\theta)\}^+]^2}{E\{W_1^2(\theta)\}}+o_p(1).$$

The asymptotic distribution of $n^{-1/2}\sum W_i(\theta)/\sqrt{E\{W_1^2(\theta)\}}$ when $f(x,\theta^*)$ is the true null distribution, is a Gaussian process with mean 0, variance 1 and autocorrelation function $\rho(\theta,\theta')$ as (5.17).

Case II: $|\theta_2-\theta^*|\le\varepsilon$.

When θ_2 is in an arbitrarily small neighbourhood of θ^*, some complications appear since $\theta_1-\theta^*$ and $\theta_2-\theta^*$ are confounded, so that $n^{1/2}(\hat\theta_1-\theta^*)^2$ is no longer negligible when compared to $n^{1/2}(\hat\theta_2-\theta^*)^2$. However, in this case, θ_1 and θ_2 can be treated equally, so that the usual quadratic approximation to the likelihood becomes possible. Since the MLE of θ_1 is consistent, in addition to $|\theta_2-\theta^*|\le\varepsilon$, we can restrict θ_1 in the following analysis to the region of $|\theta_1-\theta^*|\le\varepsilon$.

In the sequel, $\hat\pi$, $\hat\theta_1$ and $\hat\theta_2$ denote the MLE's of π, θ_1, and θ_2 within the region defined by $0\le\pi\le 1/2$, $|\theta_1-\theta^*|\le\varepsilon$ and $|\theta_2-\theta^*|\le\varepsilon$.

Again, we start with (5.24). Replacing both θ_1 and θ_2 in δ_i with θ^*, we have

$$\delta_i=m_1Y_i(\theta^*)+m_2Z_i(\theta^*)+e_{in},$$

where m_1 is the same as before, but

$$m_2=(1-\pi)(\theta_1-\theta^*)^2+\pi(\theta_2-\theta^*)^2.$$

Note that $|m_1|\le\varepsilon$ and $m_2\le\varepsilon^2$. The remainder term now becomes

$$e_{in}=(1-\pi)(\theta_1-\theta^*)^2\{Z_i(\theta_1)-Z_i(\theta^*)\}+\pi(\theta_2-\theta^*)^2\{Z_i(\theta_2)-Z_i(\theta^*)\}.$$

By integrable upper bound condition,

$$e_n=\sum_{i=1}^n e_{in}=n^{1/2}\{(1-\pi)|(\theta_1-\theta^*)|^3+\pi|(\theta_2-\theta^*)|^3\}O_p(1).$$

Thus

$$\begin{aligned} r_{1n}(\pi,\theta_1,\theta_2) &\leq 2\sum_{i=1}^{n}\{m_1Y_i(\theta^*)+m_2Z_i(\theta^*)\}-\sum_{i=1}^{n}\{m_1Y_i(\theta^*)+m_2Z_i(\theta^*)\}^2 \\ &\quad+\frac{2}{3}\sum_{i=1}^{n}\{m_1Y_i(\theta^*)+m_2Z_i(\theta^*)\}^3 \\ &\quad+n^{1/2}\{(1-\pi)|(\theta_1-\theta^*)|^3+\pi|(\theta_2-\theta^*)|^3\}O_p(1). \quad (5.35) \end{aligned}$$

Using the same argument as in Case I,

$$\frac{\sum|m_1Y_i(\theta^*)+m_2Z_i(\theta_2)|^3}{\sum\{m_1Y_i(\theta^*)+m_2Z_i(\theta_2)\}^2}\leq\max(|m_1|,|m_2|)\,O_p(1)\leq\varepsilon O_p(1).$$

Equation (5.35) reduces to

$$\begin{aligned} r_{1n}(\pi,\theta_1,\theta_2) &\leq 2\sum_{i=1}^{n}\{m_1Y_i(\theta^*)+m_2Z_i(\theta^*)\} \\ &\quad-\sum_{i=1}^{n}\{m_1Y_i(\theta^*)+m_2Z_i(\theta^*)\}^2\{1+\varepsilon O_p(1)\} \\ &\quad+n^{1/2}\{(1-\pi)|(\theta_1-\theta^*)|^3+\pi|(\theta_2-\theta^*)|^3\}O_p(1) \quad (5.36) \end{aligned}$$

Let us analyze the remainder in terms of the MLE's:

$$\begin{aligned} n^{1/2}\{(1-\hat\pi)|(\hat\theta_1-\theta^*)|^3+\hat\pi|(\hat\theta_2-\theta^*)|^3\} &= \{o_p(1)+\varepsilon O_p(1)\}n^{1/2}\hat m_2 \\ &\leq \{o_p(1)+\varepsilon O_p(1)\}(1+n\hat m_2^2) \\ &\leq \varepsilon O_p(1)+\varepsilon O_p(n\hat m_2^2). \end{aligned}$$

Consequently, in terms of the MLE's, (5.36) reduces to

$$\begin{aligned} r_{1n}(\pi,\hat\theta_1,\hat\theta_2) &\leq 2\sum_{i=1}^{n}\{\hat m_1Y_i(\theta^*)+\hat m_2Z_i(\theta^*)\} \quad (5.37) \\ &\quad-\sum_{i=1}^{n}\{\hat m_1Y_i(\theta^*)+\hat m_2Z_i(\theta^*)\}^2\{1+\varepsilon O_p(1)\}+\varepsilon O_p(1). \quad (5.38) \end{aligned}$$

Note that in the above, the term $\varepsilon O_p(n\hat m_2^2)$ has been absorbed into the quadratic

quantity. By orthogonalization, we have

$$\begin{aligned} r_{1n}(\hat{\pi},\hat{\theta}_1,\hat{\theta}_2) &\leq 2\sum_{i=1}^{n}\{\hat{t}Y_i(\theta^*)+\hat{m}_2W_i(\theta^*)\} \\ &\quad -\left\{\hat{t}^2\sum_{i=1}^{n}Y_i^2(\theta^*)+\hat{m}_2^2\sum_{i=1}^{n}W_i^2(\theta^*)\right\}\{1+\varepsilon O_p(1)\}+\varepsilon O_p(1), \end{aligned}$$

where $W_i(\theta)$ is defined in (5.16) and $\hat{t}=\hat{m}_1+\hat{m}_2h(\theta^*)$

Applying the same technique leading to (5.31), we get

$$r_{1n}(\hat{\pi},\hat{\theta}_1,\hat{\theta}_2)\leq\{1+\varepsilon O_p(1)\}^{-1}\left[\frac{\{\sum Y_i(\theta^*)\}^2}{\sum Y_i^2(\theta^*)}+\frac{[\{\sum W_i(\theta^*)\}^+]^2}{\sum W_i^2(\theta^*)}\right]+\varepsilon O_p(1).$$

By (5.22),

$$\begin{aligned} r_n(\hat{\pi},\hat{\theta}_1,\hat{\theta}_2) &\leq \frac{\varepsilon O_p(1)}{1+\varepsilon O_p(1)}\frac{\{n^{-1/2}\sum Y_i(\theta^*)\}^2}{E\{Y_1^2(\theta^*)\}} \\ &\quad +\frac{[\{n^{-1/2}\sum W_i(\theta^*)\}^+]^2}{\{1+\varepsilon O_p(1)\}E\{W_1^2(\theta^*)\}}+\varepsilon O_p(1) \\ &= \frac{[\{n^{-1/2}\sum W_i(\theta^*)\}^+]^2}{E\{W_1^2(\theta^*)\}}+\varepsilon O_p(1). \end{aligned}$$

Therefore,

$$R_n(\varepsilon;II)\leq\frac{[\{n^{-1/2}\sum W_i(\theta^*)\}^+]^2}{E\{W_1^2(\theta^*)\}}+\varepsilon O_p(1).$$

Next, let $\tilde{\theta}_2=\theta^*$, and $\tilde{\pi}$, $\tilde{\theta}_1$ be determined by

$$m_1+m_2h(\theta^*)=\frac{\sum Y_i(\theta^*)}{\sum Y_i^2(\theta^*)},\quad m_2=\frac{\{\sum W_i(\theta^*)\}^+}{\sum W_i^2(\theta^*)}\,.$$

The rest of the proof is the same as that in Case I, and we get

$$R_n(\varepsilon;II)\geq\frac{[\{n^{-1/2}\sum W_i(\theta^*)\}^+]^2}{E\{W_1^2(\theta^*)\}}+o_p(1).$$

Lemma 5.5 *Suppose that Conditions 1–5 hold. Let $R_n(\varepsilon;II)$ be the likelihood ratio statistic with restriction $|\theta_2-\theta^*|\le\varepsilon$ for any arbitrarily small $\varepsilon>0$. When $f(x,\theta^*)$ is the true null distribution, as $n\to\infty$,*

$$\frac{[\{n^{-1/2}\sum W_i(\theta^*)\}^+]^2}{E\{W_1^2(\theta^*)\}}+o_p(1)\le R_n(\varepsilon;II)\le\frac{[\{n^{-1/2}\sum W_i(\theta^*)\}^+]^2}{E\{W_1^2(\theta^*)\}}+\varepsilon O_p(1),$$

where $W_i(\theta^*)$ is defined by (5.16).

Proof of the Theorem. For any small $\varepsilon>0$, $R_n=\max\{R_n(\varepsilon;I),R_n(\varepsilon;II)\}$. By Lemmas 5.3 and 5.5,

$$R_n\le\max\left[\sup_{|\theta-\theta^*|>\varepsilon}\left\{\frac{[\{n^{-1/2}\sum W_i(\theta)\}^+]^2}{E\{W_1^2(\theta)\}}\right\},\frac{[\{n^{-1/2}\sum W_i(\theta^*)\}^+]^2}{E\{W_1^2(\theta^*)\}}+\varepsilon O_p(1)\right]$$

plus a term in $o_p(1)$, and

$$R_n\ge\max\left[\sup_{|\theta-\theta^*|>\varepsilon}\left\{\frac{[\{n^{-1/2}\sum W_i(\theta)\}^+]^2}{E\{W_1^2(\theta)\}}\right\},\frac{[\{n^{-1/2}\sum W_i(\theta^*)\}^+]^2}{E\{W_1^2(\theta^*)\}}\right]+o_p(1).$$

Since $n^{-1/2}\sum W_i(\theta)/\sqrt{E\{W_1^2(\theta)\}}$ converges to the Gaussian process $W(\theta)$, $\theta\in\Theta$, with mean 0, variance 1 and autocorrelation function $\rho(\theta,\theta')$ which is given by (5.17), the theorem follows by first letting $n\to\infty$ and then letting $\varepsilon\to0$. $\diamondsuit$